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The Problem of Coordination

A scientific theory is typically presented with laws, principles, or equations that involve terms specific to that theory:

$$s = vt + (1/2)at^2, PV = rT, F = ma, action \ = \ reaction$$

to mention only a few of the classical standbys. Although these terms are, taken literally, symbols for functions with specified mathematical character, they are often pronounced as if they were nouns already familiar before the theory's introduction: "distance", "velocity", "time", "pressure", "temperature", and so forth. That nomenclature is introduced more by way of informal commentary than explanation, and certainly does not define the theoretical terms.[1] But the choice of familiar words does signify something: they point to the sort of data to enter, and the sorts of measurements that can help to determine the values of those functions. The theory would remain a piece of pure mathematics, and not an empirical theory at all, if its terms were not linked to measurement procedures. But what is this linkage?

That question, which turns out to bring many further questions and complexities in its train, poses what was once generally known as the '*problem of coordination*'. The term had appeared in Mach's writings on mechanics and thermodynamics, was salient in the discussion of the relation between mathematical and physical geometry that extended from the nineteenth century into the twentieth, and came to special prominence through the writings of Schlick and Reichenbach when logical empiricism was beginning to break with the neo-Kantian tradition. Since then it has not been so much in use, but the problem to understand just how a scientific theory is more than its mathematical guise reappeared at every juncture in the subsequent history.

The questions *What counts as a measurement of (physical quantity) X?* and *What is (that physical quantity) X?* cannot be answered independently of each other. To echo another such realization, I am not ashamed to admit that this brings us to the famed 'hermeneutic circle' (Eco 1992: 64). We shall examine this apparent circularity by focusing on the one hand on its more abstract consideration by Reichenbach, and on the other hand the practical response in history examined by Mach and Poincaré.

Coordination: a historical context

Assigning a value, a location on a scale or in a larger space, is precisely what Ernst Mach called "coordination".[2] After a historical survey of how thermometry developed (which we will look at in more detail below), Mach began a critical discussion of how the involved choices had been made. Precisely in the midst of this critique he introduces the term "coordination":

> That number which, conformably to any chosen principle of coordination, is coordinated with a volume indication of the thermoscope, and consequently uniquely with a state of heat, is called the temperature of that state. (Mach 1986: 52)

How much would already have to have been in place before this could be offered to identify the quantity temperature? It can make sense only at a point when there is no longer any ambiguity about what counts as a thermoscope or thermometer, or any doubt about whether the volume of the thermometric substance is uniquely correlated with a 'state of heat', or about *what is meant by that*.

Mach is very clear that this is at best a late stage in the historical development. He sees clearly that it involves both idealization with respect to the stability in the instruments that count as thermoscopes and choices whose status needs to be critically examined. But some leeway for choice is brought into the open precisely in the phrase that introduces the term "coordination". He continues

> The temperature numbers are dependent on the principle of coordination, $t = f(v)$, where v is the thermoscopic volume, and, consequently, for the same state of heat they will vary greatly according to the principle adopted. (Ibid.)

As he points out, different such principles of coordination (choices of function f) were introduced. Galileo made the temperature numbers correspond to increments starting from a chosen initial volume:

$$t = N \textit{ if and only if the volume equals } v_0(1 + N\alpha)$$

while Dalton's function had the form

$$t = N \textit{if and only if the volume equals } v_0\beta^N$$

There is a simple mathematical transformation that leads us from one to the other (even if not as simple as a transformation from Fahrenheit to Celsius!), so although the two are different they are, one might say, not fundamentally different. When, however, Amontons and Lambert choose the pressure of a mass of gas of constant volume as temperature index, they are choosing a quite different principle of coordination. In the numbers assigned their method of thermometry is not markedly different from those made by Galileo's scale. But in this case, the relationship between the scales is not a matter of mathematics or logic—it is due to relations in empirical and physical contingent fact.

So what is meant by *coordination*? Just for now let us take thermoscopes and the indicated physical correlations between heating, expansion, and pressure for granted. Then the initial statement by Mach identifies the physical magnitude (parameter, quantity, observable) *temperature*. Since he identifies its value as "dependent on the principle of coordination", hence relative to a choice among coordinating principles, we must take it that only what is invariant under these choices is significant. This might seem to relegate the coordinating principles themselves to insignificance. But that is not so: in the absence of any such coordinating principle, attributions of *temperature* would have no empirical content at all, no reference in what is already established as measurable. To put it briefly: on the one hand, coordination requires for its possibility some recognized empirical regularities, while on the other hand it is required for the new theoretical assertions to have any empirical content at all.

The term "coordination" was central in Schlick's *Algemeine Erkenntnislehre*, where the account of the process of cognition is based on idea of coordination (*Zuordnung*).[3] While I will ignore his generalization as only tangentially relevant to our subject, its historical role underlines the impact of this problem on the philosophy of science that was then developing.

Here Poincaré (whose analysis of this problem for time measurement we will look into below) had been a central figure, and almost every discussion had to orient itself with respect to his conventionalism. Einstein took an active part, and in fact uses the terminology of coordination quite explicitly in his 1921 essay, "Geometry and Experience":

> It is clear that the system of concepts of axiomatic geometry alone cannot make any assertion as to the behavior of those objects of reality which we designate as practically rigid bodies. To be able to make such assertions, geometry must be stripped of its merely logical-formal character by the coordination of experienceable objects of reality with the empty conceptual schemata of axiomatic geometry.[4]

But the radical, foundational taking by the horns we find in Reichenbach's 1920 book, *The Theory of Relativity and A Priori Knowledge*. There we see him landing in serious difficulty with the question of how terms in the theory of space, time, and motion—whose primary reference is to mathematical objects—can be linked to items or aspects of the physical. The reason for his difficulties lie precisely in the radicalism of his philosophical approach.

Coordination and Reichenbach's problem

Mach's discussion had encountered no difficulty in principle because he stated his principles of coordination in a context where certain other physical parameters and their measurement were assumed to be given. The measurement of length or volume and volume change, for example, is simply taken for granted in the passage I quoted.

In 1920 Reichenbach, on the other hand, realizing that he is addressing the most fundamental level of physical theory, allows himself no such luxury. He wants to find a general coordination of mathematical spaces and their structure with physical relations. In his later *Philosophy of Space and Time* he gives as first example how units of length can receive their coordination: "a meter is the forty-millionth part of the circumference of the earth"(Reichenbach 1958: 15). But the recurring example of the concept of a straight line, or more generally a geodesic, as having as physical correlate a light ray, or the path of a freely falling body, is not equally easy to understand. Nor is the idea that congruence relations have as correlate coincidence with transported rigid bodies.[5] In these examples it is quite unclear how to identify the physical correlate without using any geometric or kinematic terms. How are we to describe rigidity or free

fall without using the language of geometry, or of mathematical physics in general? In the example that most preoccupied him about the theory of relativity, non-simultaneity is to be thus related to non-connectability through any signals, whether by light emission and reflection or by material transport. Not only the modal character of "connectable" but also the required identity-over-time (genidentity) of the material are as puzzlingly theoretical as any of the terms in physical geometry. And so we find him perplexed. He writes in 1920:[6]

> It is characteristic of modern *physics* to represent all processes in terms of *mathematical* equations. But the close connection between the two sciences must not blur their essential difference. (Reichenbach 1965: 34)

> The mathematical object of knowledge is uniquely determined by the axioms and definitions of mathematics. (Ibid.)

> The *physical object* cannot be determined by axioms and definitions. It is a thing of the real world, not an object of the logical world of mathematics. Offhand it looks as if the method of representing physical events by mathematical equations is the same as that of mathematics. Physics has developed the method of defining one magnitude in terms of others by relating them to more and more general magnitudes and by ultimately arriving at "axioms", that is, the fundamental equations of physics. Yet what is obtained in this fashion is just a system of mathematical relations. What is lacking in such system is a statement regarding the significance of physics, the assertion that the system of equations is *true for reality*. (Ibid.: 36)

So how can empirical significance be achieved? The examples he has in mind, as we just saw, are the use of rigid bodies as choice to set the relation of spatial congruence (in effect, to measure length), the choice of a light ray path in vacuo as *physical correlate* for geodesics, the choice of a certain periodic process as setting the unit of time. Question: how are these physical correlates to be identified without use of geometric or kinematic terms?

Look back to Mach's examples of Galileo's and Dalton's choices for temperature scaling. These choices are expressed in the above equations in which one variable is set equal to a function of another variable. So is the choice in question simply a choice of a *function*? A function relating what to what? Isn't a function a mathematical object itself, defined in terms of a relation between *mathematical* objects? So Reichenbach writes:

> The coördination performed in a physical proposition is very peculiar. It differs distinctly from other kinds of coördination. For example, if two sets of points are given, we establish a correspondence between them by coördinating to every point of one set a point of the other set. For this purpose, *the elements of each set must be defined*; that is, for each element there must exist another definition in addition to that which determines the coördination to the other set. Such definitions are lacking on one side of the coördination dealing with the cognition of reality. Although the equations, that is, the conceptual side of the coördination, are uniquely defined, the "real" is not. On the contrary, the "real" is defined by coördination to the equations. (Ibid.: 37–8; my italics)[7]

Here Reichenbach was imagining, and discounting, the following naïve sort of reply:

> *what is called for is simply a function, a mapping, between mathematical objects and physical objects or processes—what is puzzling about that?*

The reason he discounts it is because to define a function we need to have the domain and range identified first—and the question at issue was precisely how that can be done without presupposing that we already have a physical-mathematical relation on hand.

So what does Reichenbach mean, when he seems to point to a solution with the words "On the contrary, the "real" is defined by coordination to the equations"? We are here at the historical point where the split between the neo-Kantian tradition and logical empiricism begins, at least in Reichenbach's intellectual life.[8] In this early book, Reichenbach argues that at least one essential element of the Kantian a priori can still be maintained. So to effect the necessary coordination of abstract mathematical structure to concrete empirical reality he posited a special class of mathematical–physical principles—"coordinating principles" or "axioms of coordination"—whose role is precisely to insure those conditions of possibility. These principles, Reichenbach argues at that point, are to be taken as given or imposed a priori, and so to be sharply distinguished from mere empirical laws ('axioms of connection').[9] In this respect then he follows Kant's original conception, though he adds the innovation that the a priori is historically *relative*, the coordinating principles must be changed as the sciences develop.

As we should guess from Mach's example, Reichenbach can coordinate a mathematical representation with physical objects, events, and processes only in a context where something is already given that will make that

possible. Thus we must doubt whether Reichenbach really did, by this reconception of the a priori as historically relative, provide something given-in-context that makes it possible to link mathematical objects to physical magnitudes. How could principles, or having principles, provide the conditions in which we can relate something mathematical to something physical?[10] What exactly *are* the required conditions—isn't that the question to be answered before we can think about what would make those conditions possible?

In fact, it seems that Reichenbach himself was not able to maintain this reaction to his problem of coordination. At a somewhat later point in his writings he offers "coordinative definitions" instead of axioms of coordination. This was, I think, definitely not meant to be just a verbal change: he means to trade at that point on the conventionality of definition, the possibility to bring the definition—and therefore, supposedly, the coordination—into being by an act of stipulation. At the same time he means to be still actively inquiring into the empirical conditions under which such definitions can play the requisite role. What he did not do is change the ahistorical setting of his problem: the coordination is still apparently to be conceived of as possible in the absence of any previous such coordination. But is that possible at all? We have to ask more or less the same question again as before: how can such coordinative definitions be meaningfully introduced except in a historical context where there are some prior coordinations already in place?[11] I submit that they cannot.

The problem of coordination reconceived

How should we see the problem of coordination now? Not as a problem so posed as to preclude its own solution! Nor does it seem that we get very far by posing it as a problem that is solved by choosing a particular object to be designated as standard of unit length, or light-ray paths as physical correlate for geodesics.

Coordination must determine how measurement can establish a value for what is measured. But that it can do so appears to presuppose an understanding both of the measurement procedure and of what is measured—that is, of the terms between which that relationship is established through coordination. So the prior question at issue is precisely:

What is measured? In the assertion "the length of A was measured", or "the temperature of B was measured" or "the duration of C was measured", what exactly is the referent of the subject term?

This question will look different depending on where we take our vantage point. There are actually two distinct vantage points common in the philosophy of science literature—not just about measurement but about experiments, models, theories, . . . But the divide becomes especially salient in the ways measurement is discussed.

- We have no problem identifying what is measured by a particular procedure if we look at it 'from above', that is, *in retrospect*, from within a theory that is already stable and established, and deals with that physical magnitude.

Looking at the matter from this vantage point we say that what was being measured all along, however well or poorly, in the centuries leading up to that achievement, was precisely what we say it is now.

- Nor do we have a problem in our discussion of measurement if we look at it 'from within', that is, if we inquire into the introduction of measurement procedures *at historical stages where there were already measuring procedures* for certain other physical magnitudes *taken as given*.

The examples of Mach on temperature and Poincaré on time measurement, that we will look at below, illustrate this very well.

It is only if we try to understand coordination without locating ourselves either in the ahistorical 'above' or the historical 'within', but pretend to a 'view from nowhere', that we set ourselves an impossible problem—or fall into metaphysical metaphors.

If in our retrospective view today we want to describe the historical process, we can of course only speak our own language, the language we have today. And today it is a truism that

> an item has temperature T if the stable procedure that we now designate as temperature measurement, when applied to this item, has outcome value T.

So if today we have to say what the earlier activities we focus on—such as Galileo constructing a sort of thermometer—were all about, and have to answer what was the intended referent, while measurement procedures

and theories were still evolving, our answer had better be coherent with what we *now* take that referent to be. So we can only say:

> all along they were referring to the magnitude identifiable as measured by the procedures that would eventually be accepted as measuring it.

So yes, it makes sense for us to say that Galileo's activity was about measuring temperature, although we have to add immediately (speaking in our own voice, using our own words) that his apparatus did not really do that, that there was a mistake involved in his set-up.

Notice how this answer echoes Reichenbach's statement that what is measured, and thus through measurement coordinated with a theoretical concept, is not defined independently of that coordination, but defined by the coordination. At first blush this formulation too may sound circular, even viciously circular, if not debilitatingly anthropocentric or egocentric. But it is not. What would make it viciously circular is only Reichenbach's tacit assumption that what was being done could in principle be done without any historically prior achievements along such lines. But to understand this properly we must follow the process of coordinating both in its general form and in specific examples, to show how the joint evolution of measurement and theory can happen and can come to a stable resting point.

What is measured?

Can we identify the parameters that are measured by means of the measuring procedure itself alone? Well, what does "alone" mean here? Does it mean "with no theoretical background in which to operate"? But with no theoretical background, very few procedures classify as measurement procedures. For example, as Mach and later Poincaré emphasized, procedures for measuring mass presuppose something about the items to which they are applied, e.g. that they are subject to the law that action = reaction.[12] This is precisely why Mach, and later Cassirer, Schlick, and Reichenbach, in their different ways, concerned themselves so much with the problem of coordination.[13] But it is also through measurement results that theories gain empirical support, so there must be some trade-off, or some subtleties to how empirical support can be gained, if we are not to find ourselves lost in a circular argument.

The idea that the parameters can be identified simply through the measurement procedures was behind Bridgman's *operationalism*, and we often see its strong appeal still in scientific writings. But the specification of the measured parameter *even* in terms of the (final, stable) measuring procedure can never amount to a complete identification or definition. For the procedure will classify only when it is actually applied, and even then only establish order among the items to which it is applied, at the times when it is applied.[14] To try and stick with an 'operationalist' view of what is measured would land us in the absurdity that a process or object has no character at all while it is not being measured.

In practice a theory eventually emerges which encompasses the measurement procedure itself as well as the items measured, and provides the coordination. Thus in the case of temperature, the kinetic theory which was developed after thermometry had been developing for two centuries, provided the parameter which then was identified as precisely what was measured by the thermometer. So a definite identification, a complete definition, of the measured parameter is possible but only through, at the hands of, and relative to the theory offered and finally accepted to account for the stability of the measurement procedure.[15] We need to see this development in its historical perspective, and to recognize and distinguish two facts:

(a) an *empirical* fact that has been discovered: namely, the very stability in the procedures found in this historical development, and the reliability of the predictions concerning these and their correlation with other measurement procedures derived from the mature theory in which they are now theoretically embedded.

(b) a *historical* fact: namely, that choices in the development of these measuring procedures went hand in hand with the development of the theory in question, so that we cannot identify an aspect of nature that is measured if we refer only to those empirical procedures without using concepts provided by the theory.[16] What now counts as simple passive measurement is a hard-won achievement.[17]

What were the choices involved? What epistemic status can we assign to these choices? We will not gain much clarity if we continue simply in the current abstract vein. We need to look at the actual history to get any grip on this at all.

Mach on the history of the thermometer[18]

As Mach emphasized, it was already a matter of actual practice, before modern science turned its attention to the question of a quantitative measure, to evaluate bodies as hotter or colder or of equal temperature on the basis of common experience. But judgments based on perception do not yield an unambiguous ordering—as was clear from many common examples. Iron and wood never feel equally hot or equally cold no matter how long they are left undisturbed in contact, for example. Evaluations of hotter and colder tend to be reversed if the perceptions are reversed in order. As Poincaré would later say for time measurement, psychologists could ignore this but not the physicist: some grading procedure with results that could fit into a single scale was desirable.

To pursue this desideratum is already a choice. Almost from the very beginning, the aim was to establish a numerical scale. The ideas of *hot, cold, as hot as,* and *hotter than* were to be transformed into the idea of a quantity with the same general structure as height or volume or speed.

The air thermometer

The word "thermometer" came into use in the mid-seventeenth century. In the first half of that century there had already been widespread use of instruments that responded to temperature changes, based on the expansion and contraction of air. Galileo is sometimes credited with the invention; certainly there were examples in Italy in his time. Retrospectively they are called "air thermometers". A flask with a long neck is vertically inserted into a container of water, with the opening downward. As the flask becomes colder, the water in the neck rises. Similarly, when the apparatus becomes warmer, the water goes down. The differences in water height were at first measured with compasses, but then divisions were marked with numbers attached. Records were made of how low and how high the water would be in winter and summer, or during the night and the day.

The very set-up cries out for a simple numerical scale: how high is the column of water in the tube? Changes in that height indicate precisely the change in volume of the contained air. The guess is that the air expands with increase in temperature and contracts as it becomes colder. This guess does not presuppose that we already know how to measure how much

warmer or colder it gets: the rough and ready comparison by the senses will do for that. But if the change in height of the water column, correlated with the volume of enclosed air, can be used as indicating *how much* hotter or colder the air became, then we will have a numerical scale induced by the most basic, most venerable form of already familiar measurement: *length*.

Is there an empirical question involved in this choice? What cannot be ascertained without the use of a good thermometer is whether the change in volume is *quantitatively proportional* to the change in temperature. There will certainly be empirical questions to come, but this is not one. *Prior to the construction of a thermometer, there is no thermometer to settle that question*! So here the scientist confronts a choice, one that may be provisional to start, with hope of vindication of some sort or other later on: to use changes in volume as quantitative measure of how much hotter or colder one thing is than another.

There are a number of drawbacks in this arrangement, if we try to think about it in current terms as measuring temperature. The water and the glass are also relevantly affected by temperature changes, even if not to the same degree as the air. The initial temperature of the apparatus will also play a role. But most important from that point of view is the effect on the water level of changes in atmospheric pressure—the instrument is, *we might say now*, part thermometer and part barometer. This could not even be realized or taken into account until the notion of atmospheric pressure was in play. The history of the barometer began during the same century, and like that of the thermometer was initially beset by theoretical disputes about the possibility of a vacuum. Hence the problem of effects of atmospheric pressure on the air thermometer did not appear till it could appear. The problem that the air thermometer was actually a mixture of thermometer and barometer was in fact pointed out by Blaise Pascal in mid-seventeenth century, after his barometric experiment on Puy-de-Dôme.

The liquid thermometer

How was this problem to be avoided; how could one have an instrument to gauge temperature differences that was not "part" barometer? In the second half of the century the use of the liquid thermometer became common: a liquid in a glass tube, filled completely when at very high temperature and hermetically closed. When it cools to room temperature, the liquid no

longer fills the whole tube—arguably, and at the time contentiously, one now witnesses the creation of a vacuum. (Questions on that score were crucial then; I shall ignore them here.) A common form had its length divided into equal parts. Detailed instructions recorded at the Accademia del Cimento in Florence, specified that in a good construction snow or ice will not make the liquid column descend below the 20 degree mark, nor the hottest day dilate it beyond 80 degrees.[19] Since the tube is entirely closed, this arrangement eliminates the effect of atmospheric pressure, though not the effects of temperature changes on the liquid or the glass.[20]

The assumption or choice of definition that differences in temperature are proportional to resulting differences in volume continues, though now instantiated to liquids. Two obvious questions: do different liquids agree with each other in this respect? and do they indeed always expand when, by other common criteria, they are becoming warmer? These questions did not receive very satisfactory answers. The early makers of liquid thermometers used quite a variety of different liquids. Water was not a good candidate, since its volume does not keep contracting as temperature drops toward freezing.[21] None of the liquids is of any use below its freezing point anyway. But in the regions well above their freezing point it was at first thought that they all expanded and contracted in the same proportion when put in the same environment controlled for effects other than temperature variation—an appearance which in turn was refuted by Dalton for the case of high temperatures.[22]

The results are thus sensitive to the choice of liquid, partly because their response to temperature changes varies at very low and very high temperatures in a certain range—and the liquid state does not persist below or above a certain point. Even in 'good' ranges, the expansion in volume is not proportional for different liquids. Dalton wrote, a good century into this evolution, that liquids were found to expand unequally, "but no two of them alike. Mercury has appeared to have the least variation".[23] But even for the otherwise very good choice of mercury, the effects on the glass tube are not negligible in comparison to the effects on the liquid. The expansion of mercury is only seven times that of glass.

The gas law and kinetic theory

These findings occasioned a return to the gas thermometer, especially in view of the discovery and refinement of what is now often known

as the ideal gas law (Boyle's Law, the Boyle-Mariotte Law, the Law of Charles and Gay Lussac, Charles's Law—a multiplicity of names partly due to nationalist partiality), $PV = rT$, with r a constant characterizing the gas. If pressure is kept constant, the volume varies proportionally with the temperature, for all gas in the same way—conversely, if the volume is constant, the change in pressure is proportionate to the change in temperature. Obviously the experimental findings could at best hint at this relationship before thermometry had been stabilized. But of gases, unlike of liquids, it could not be said that they expand unequally "but no two of them alike".

Guillaume Amontons pointed out, in effect, that if a law of this form is correct, then the temperature scale can become a *ratio scale*: it will have an 'absolute zero', marked by the theoretical condition of zero pressure.[24]

Is this law correct? To begin its correctness is supported by experimental findings, made with thermometers of the previous generation, accepted as a rough guide while recognized as inadequate. Then, as confidence in the kinetic theory increases, *theory cuts the Gordian knot* with respect to all questions about how the scale is to be fixed. The gas law is supported by its incorporation in the kinetic theory—and refined as well, so that it also becomes itself qualified there as having been only an empirical approximation to what really happens. But the scale is now anchored in prior scales for mechanical parameters. Once T is identified with the mean of a mechanical parameter, the postulated mechanical behavior—"the type of motion we call heat", in Clausius's incomparable phrase—is now the referent of what is measured.

We can also put it the other way around: the parameter, identified by the eventually stabilized procedures for its measurement, is now classified by the theory as one aspect of the logical space that the theory provides for location of items in its domain.

The history of thermometry did not stop here, but this much will suffice to make the important points concerning coordination.

Choices and conventions

The new type of instruments that emerged in this evolution, before the kinetic theory rewrote the story of what has happened, were selected as correct for thermometry in a process that involved three main choices. Mach's history of the subject emphasized perspicuously the sense and

extent of these choices involved in how the scientists arrived at their results.

The first choice, which we have been emphasizing from the beginning, concerns what is to be taken as representing the quantitative change in temperature. The choice of change in volume passes the initial test of rough agreement with the admittedly imprecise deliverances of experience in familiar circumstances. The vindication of such an initial choice in the future is of course hostage to fortune. The second is the choice of thermoscopic substance, to answer "change of volume of *what*?" And the third is just what above, at the outset of our discussion, we saw Mach calling the "principle of coordination" for assigning numbers, that is, for setting the scale. (op. cit: ch. II, sections 11–12).

There are coherence constraints on the choice of thermoscopic substance. We have seen that if it had been decided to stay with liquids, there would have been quite a lot of arbitrariness in this choice, first of all because they do not appear to expand at the same rate even in the same range. Secondly, different liquids would have to be chosen for different ranges, as they near their different freezing or boiling temperatures. The gas law saved the day: it seemed that different gases expand at the same rate as each other under the same conditions. Again there are limitations, since real gases too could liquefy and solidify at different temperatures.

The last choice, to set the scale, may now seem trivial. But it is not. The significant choice involved is best seen in the discussion of whether there is an absolute zero, a minimum to the temperature scale. The main convention, historically, goes back as far as Galileo: to take the melting point of ice and the boiling point of water as selecting fixed points on the scale. We have already seen that the change in volume of water will not be a convenient one overall. Using the gas law, for a particular gas, we can define the temperature scale either in terms of that for volume at a constant pressure or in terms of that for pressure at a constant volume. So Amontons proposed the pressure p of a mass of gas at constant volume, and chose as reference point the pressure p_0 at the melting point of ice. Then the temperature scale can be defined with an arbitrary scaling constant k as $t = kp/p_0$. The numbers k and the pressure scale can be chosen so that the difference between the temperature of melting ice and that of boiling water is 100, and there is an absolute zero of temperature for $p = 0$, whether this can be attained in nature or not.

This is a very natural way to proceed, but if it is thought that the lower bound is a fiction, one could choose a different principle of coordination, not bounded below any more than bounded above. Mach points out that Dalton had in effect chosen such a scale, though of course that is not the one that caught on.

The choice is settled later by the absorption of the entire subject of inquiry into physical theory. In the kinetic theory the magnitude with which temperature is identified has an absolute zero point. But it is well to note that this was not part of the empirical facts guiding the evolution of thermometry, though the kinetic theory was developed in a way that was accountable to that evolution of empirical procedures.

The very simple operations we now perform with thermometers are an obvious example of 'passive' measurement—in our theories we have the interaction as well as the object and apparatus completely classified, 'well understood'. This stage is an *achievement*. The story of thermometry is a success story, and such success was by no means guaranteed a priori. The conditions of possibility for such a successful coordination—here we are at Reichenbach's question—we have now seen to involve empirical regularities that are contingent and choices subject to conditions of coherence.

Poincaré's analysis of time measurement

The second half of Poincaré's essay "The measure of time" is the more famous because of its connection with special relativity.[25] But I will concentrate here on the first half, where Poincaré begins with the problem that we do not and cannot have a direct intuition of the equality of successive time intervals (equality of duration of successive processes).[26] This is not a psychological point. Two successive periods of a clock cannot be compared by placing them temporally side by side, that is why direct perception can't verify whether they lasted equally long. This had been a recurring topic of discussion; Poincaré mentions Andrade but across the Channel too it was a topic among the British Idealists and for the young Russell.[27] Psychologists may ignore this, Poincaré writes, but the physicist cannot.

In the case of two sticks we can check to see whether they are equally long (at a given time) by placing them side by side; that is we can check spatial

congruence (at that time) by an operation that effects spatial coincidence (at that time). We can check whether two clocks run in synchrony during a certain interval if we place them in spatial coincidence. These procedures do not suffice for checking whether two sticks distant from each other in time or space are of equal length, nor whether distant clocks are running in tandem, nor whether a clock's rate in one time interval is the same as some clock's rate in a disjoint time interval. But in physics, criteria for spatial and temporal congruence are needed. Poincaré is concentrating on this need.

What measures duration is a clock, and physics needs a type or class of processes that will play the role of standard clocks. What type or class to choose? One answer might be: the ones that really measure time, that is, mark out equal intervals for processes that *really* take equally long. While certain philosophers or scientists might count this demand as intelligible, it must be admitted that there could be no experimental test to check on it.[28] We cannot compare two successive processes with respect to duration except with a clock; but clocks present successive processes that are meant to be equal in duration. This is similar to Mach's point about thermometry: whether the melting of ice always happens at the same temperature, or the volume of a substance expands in proportion to temperature increase, can be checked only with something functioning as a thermometer—and thus cannot be ascertained in order to check whether thermometers are 'mirroring' temperature.

So Poincaré presents the choice as a decision: to begin, he says, it was decided to "admit by definition" that successive periods of a pendulum had equal duration. This is not a historical claim about conscious or explicit decisions, let alone about the actual practices of time reckoning before the modern era. The clock mechanism driven by a descending weight was in use before Christiaan Huygens and Galileo independently invented the pendulum clock. Poincaré is writing a "just so" story, a way to understand how choice not subject to direct empirical check is involved, no matter which mechanism is settled on to begin. Some choice—which may be subject to equally chosen constraints but not to empirical verification—is needed to provide an initial starting point.

Coherence conditions

But it certainly is the sort of choice that is also subject to strong coherence constraints, which are not a matter of choice. The class of processes chosen

as initial standard for equal duration must be such that *if two are in spatial coincidence, they run at the same rate, they run in synchrony*. The decision to choose the pendulum, in such bare form, cannot be sustainable in practice, precisely because it does not have this coherence. Different pendulums will disagree due to alteration in temperature, friction, barometric pressure, etc., and possibly beside these due to at that time unknown or uncontrollable factors. Comparing the behavior of different pendulums does not display stable synchrony. In 1672 Richer took a pendulum clock from Paris, latitude 49° North, to Cayenne in French Guyana, latitude 5° North.[29] He found that he had to change the length of the pendulum to make it agree with standard time reckoning, precisely because as Huygens and later Newton were able to explain, the force of gravity is not the same everywhere on earth. So the next development, according to Poincaré, was to correct mechanical clocks such as pendulums by the sidereal clock, which is based on the passing of a star across the meridian—the unit then being in effect the duration of one rotation of the earth. This too is a decision, and this time it was made explicitly.[30]

Yet this adoption of the sidereal clock was subject to a similar objection, if in a somewhat more theoretical guise: the movement of the seas will act like a brake, slowing down the rotation of the earth. Far from being a merely academic or skeptical doubt, this point was used to explain an apparent acceleration in the movement of the moon.

Poincaré wishes to reveal by these examples two problems that arise in developing a measurement procedure for duration. The first is the initial one, illustrated with the pendulum: we cannot place successive processes side by side so as to check whether their endpoints coincide in time. So there is no *independent* means for checking whether successive stages of a single process are of equal duration: the question makes sense only after we have accepted one such process as 'running evenly'.[31] As said above for temperature,

> duration = the magnitude identifiable as measured by the clocks that would eventually be accepted as measuring it.

This insight requires us to move all our concern to *coherence constraints* that put bounds on what we can choose to regard as 'running evenly'.

In another context, to illustrate that definitions are not totally arbitrary but in general have factual presuppositions, Poincaré gave the example of

the *light-clock*. Suppose we design a device in which light is emitted at one end of a meter-long path, reflected at the end, and absorbed again at the starting point. The interval between emission and absorption could be taken as the unit of duration. Now we carry this device and many faithful copies all over the place to be used as clocks. Any objection to this? Yes: in nineteenth-century physics it is entailed that these clocks will not give consistent results, because even rotating them with respect to each other will result in violations of synchrony. No: after the Michelson-Morley experiments, or more accurately, after Einstein's special theory of relativity was accepted, these light-clocks are perfectly acceptable in principle: there is no necessary incoherence.

The second problem is brought out by the astronomers' rejection of the equality of sidereal days. Their rationale for this is an application of Newtonian physics to the earth and its seas. The application of this physics presupposes that the reference for its time parameter has been fixed. Fixing that parameter by taking sidereal days to be equal runs into trouble precisely with the theory's representation of the very processes that define the sidereal day. So Poincaré diagnoses the astronomers' practice as follows: *they define duration in such a way as to ensure that Newton's laws come out correct*.[32] But this decision, that the time parameter should be fixed so that Newton's laws will come out true, does not suffice by itself! Newton's laws could be re-expressed in terms of different ways of measuring time; they just would not have the same simple form.

Suppose for example that with a certain adopted standard of measurement and its time scale t, these laws are satisfied. Now adopt a new scale T, which is a logarithmic function of t. It would be silly to reflect that Newton's laws as originally stated are not satisfied if t is replaced by T: the thing to do is to translate Newton's laws by means of the transformation of t into T. The result will be an empirically adequate theory if the original was—it will just be not nearly as simple or user-friendly. So in rejecting the sidereal day as time unit, the astronomers, according to Poincaré adopted not just the above criterion but the following more complete one: *Time is to be defined in such a way that the equations of mechanics will be as simple as possible*.[33]

Coherent rivalry and choice

This quick conclusion we'll have to look at critically in a moment, for surely Poincaré is going a bit fast here! Simplicity is perhaps only a code

here for a cluster of pragmatic desiderata for our theories. But we can add that the issue became concrete later on due to a proposition advanced in cosmology by Arthur Milne in the 1930s.[34]

By that time the corrected astronomical time reckoning that had become standard had been supplemented at least at the theoretical level with atomic clocks. It was part of the accepted theory that atoms of each chemical element and compound absorb and emit electromagnetic radiation at their own characteristic frequencies. As reckoned by the sidereal standard, these resonances had been taken to be inherently stable over time and space. In Milne's time there was as yet no practical application—the first atomic clock was not built till 1949. Recall now that there are coherence constraints: the class of "standard" clocks must be such that whenever two are in coincidence, they run at the same rate, they run in synchrony. In principle there could be two such classes, disjoint from each other, and not combinable in a way that satisfies this constraint. Milne proposed that this is indeed the case, and that astronomical (sidereal) clocks run differently from atomic clocks. The disparity is negligible outside cosmology. The relation between them, he submitted, was an exponential function, so that the two give us entirely different pictures of cosmic history: for the one, the past is finite and for the other it is infinite.

Although a lively topic at one stage in the development of cosmology, the idea did not have any continued impact. What it illustrates and is important, though, is the point of principle already made by Poincaré. If the class of measurement interactions (of one sort or another) cannot have an "if and only if" definition *but only be required to satisfy certain coherence conditions*, then the possibility of such disparities (and hence leeway for convention or choice) is entailed.

Einstein's critique; an example with length

In his largely sympathetic discussion of Poincaré's views, Einstein famously added one telling criticism.[35] While choice is involved in setting up the principles of coordination, this sort of choice can enter at many different points, and therefore it is not correct to single out any one element as a stipulation, definition, or convention.

This becomes clearer when we consider the development of measurement processes for more than just one observable. Without some element of choice or convention, coordination is inevitably circular. So far I have,

in all the examples, taken length measurement for granted. But suppose we consider length in the same way as we considered temperature and time measurement, where the principles of coordination presupposed length measurement as given. In order to ensure standards of length comparison that would remain applicable across time, another such history of coordination unfolded. One telling example occurred early on in a proposal by Huygens. After his invention of the pendulum clock, and his appreciation of how its period is related to latitude, he proposed that the unit of length should be one third of the length of a pendulum with a one second period at the latitude of Paris—not the English foot, not the *pied du Roi*, which was 16/15 of the English foot, but the *pes horarius*. Here time measurement is presupposed for the coordination of length.

While this proposal did not prove practical, we see a similar entanglement of several principles of coordination at the most familiarly known stage: the construction of the standard meter embodied in the platinum bar to be kept in the French State Archives. Iron copies were distributed to other countries, and needed to be sent to France every ten years to be compared with the *primus inter pares* copy kept in Sèvres. So we see two decisions installed as definitive: that such metal bars remain congruent to themselves under transport, and so provide a standard for congruence with respect to length, and that a particular such bar defines the unit of length. But what precisely was the convention? It was decided at the Convention of the Meter in 1875, and was precisely this: the meter is defined to be the distance between the midpoints of the ends of the *mètre des archives* at the temperature of melting ice. But is the temperature of melting ice always the same? The convention presupposes temperature measurement, just as the conventions for temperature measurement presuppose length measurement. The circle can be broken by entering a choice at one point or another, but there is no specially privileged point for entering a choice.

Thus Einstein separated Poincaré's rather simplistic pointing to specific 'conventions' or 'definitions' from the real insight into the inevitable holism in the jointly constructed theory and coordination. As Ryckman observes, Einstein's admission of Poincaré's viewpoint "is not a concession to conventionalism, that is to the freedom to choose any geometry we like, but to the inevitable epistemological holism of a theory in principle capable of explaining its own measurement appliances, and so its ties to observation" (Ryckman 2005: 64). But this holism will isolate the theory

from what it is about unless the coordination is effective. This can be come about in many different ways, with an element of choice appearing at different points—just not without entering any choices at all. Let's summarize it this way: any way you slice the cake into equal pieces, some choice is involved, but you can't single out one slice as the unique result of choice and the others as compelled.

Poincaré was simplistic also in his emphasis on simplicity as playing a decisive role in the choices made along the way. There are many desiderata for theories in science. The bottom line is certainly and only empirical adequacy. Pragmatic considerations such as simplicity provide desiderata as well, but they are various and diverse. There is a practical need to align the continuation of physical theory in form with what is in place already. Thus the desire for explanation, or for conformity to certain favored patterns such as common cause models, could outweigh simplicity. None of this qualifies the important point, that the crucial and necessary coherence constraints on what qualifies something to play the role of physical correlate to measurement leave much leeway—so that choice is inevitably involved—and that there is no independent recourse beyond coherence.

To be sure, this requirement of coherence is not simply one for logical consistency. Whether a sort of mechanism can be used to define the family of standard clocks depends on empirical regularities that may or may not obtain. The central coherence condition on the family of standard clocks, recall, was that *if two are in coincidence, they run at the same rate, they run in synchrony.* That is a matter of empirical fact. But if two sorts of mechanism satisfy this condition, there is no matter of fact as to which runs evenly, and a choice or convention alone can decide on one of them.

There is a natural drive once the role of such choices is recognized, to reformulate theory in a format independent of those choices. The most salient example is the ever greater generality with respect to coordinatization, which had begun earlier in mechanics, and the later extreme freedom in this respect aimed for in general relativity and even more in the space-time theories of Weyl and Eddington.[36] The other great example is the advent of group-theoretic formulations, in which invariance under as wide a class of transformations as possible is the guiding aim. But eventually these theoretical representations, which abstract from applicable models to that extent, have to be brought back into applicable form, and there choices determining the particularity of that form come back into their own.

Observables coordinated: two morals

There are at least two morals to be drawn here, and the first is an anti-foundational moral we can draw from Reichenbach's impasse. The rules or principles of coordination that can be introduced to define particular sorts of measurement cannot even be formulated *except in a context* where some forms of measurement are already accepted and in place. In the use of astrolabe and ruler to measure distant lengths the local measurement of lengths and angles was taken for granted. In the construction of thermometry, the measurement of volume and weight, and later pressure, was already in place. In the measurement of duration, measurement determining pendulum periods, and later on astronomical alignments, was presupposed as already possible throughout. *There is no presuppositionless starting point for coordination*. We are, to adapt Neurath's metaphor, sailors engaged in a continuing construction, renovation, alteration, and repair of the ways in which we measure our ship while already at sea.

Reichenbach, I would say, erred on the side of philosophical prudence, precision, and caution, because he realized the crucial importance of bestowing genuine empirical content on the equations in a theoretical description. A case of philosophical scruples . . . but scruples, as the Church emphasizes, are also a source of error.[37]

There is a second moral. The opposite error is more tempting, and less excusable. It is the error that Reichenbach described and from which he rightly shied away (even if too far). For, as he saw, the natural temptation in response to the problem of coordination is simply to impose a parallel vocabulary and declare victory. One might say, for example,

> a point in Minkowski space corresponds to a real or physical space-time point, which is a compact convex part of the WORLD with zero measure.[38]

Reichenbach would be quite right to retort that this parlance is meaningful only *after* the problem of coordination is solved. It cannot be a solution! For this "solution" begins by thinking of the WORLD as a structure, a set of elements with certain relations between those elements, described precisely in the terms we use in geometry. So it takes for granted that we can represent the WORLD mathematically. And what is that, if not to say

that there is a certain 'representing' relationship between the WORLD and some mathematical structure?—thus assuming precisely what was to be illuminated.[39]

Reichenbach offers another familiar example as illustration. Boyle's Law is an equation: PV = rT. To give this physical meaning, its terms must be 'coordinated' with physical quantities, pressure, volume, and temperature of gases. But what is, for example, the temperature of a gas? It changes with time, and differs from one body of gas to another, so isn't it a function that maps bodies of gas and times into the set of real numbers? But if I say that, am I not already taking for granted that I have at my disposal a description of bodies of gas and times? Could this be a description that does not use terms relating to spatial boundaries and numerical comparison? I can try to cut the Gordian knot by saying 'There exist physical quantities P*, V*, and T* which pertain to bodies of gas, and when "P", "V", and "T" refer to these respectively then "PV = rT" is true'. That is to do just what I said above: to create a parallel vocabulary and declare victory. But it is an empty victory.[40]

What that sort of naïve response ignores is that there is no independent epistemic access to the parameters to be measured—no access independent of measurement, that is. How could one decide, before a detailed theory is in place, whether or not the changing height of a column of mercury mirrors the changing temperature, except by use of a thermometer?[41] So, before thermometry has been established, how could one decide that at all? The answer is that the question is absurd.[42] A metaphysician may postulate that there are such physical magnitudes P*, V*, and T*, but s/he is building castles in the air if the next step involves some god-like vision or ontological telescope to compare their values with what our instruments show.

Instead, as Mach's and Poincaré's analyses show, measurement practice and theory evolve together in a thoroughly entangled way. Somewhat hesitantly one might say that the measured parameter—or at the very least, its concept—is *constituted* in the course of this historical development.[43] Choices are made, and once made may encounter resistance, whether in experiment or in theory-writing or (more usually) in combination of the two—or else vindicated by smooth progress on both fronts. Mach oversimplified by keeping theory as much in the background as possible, and Poincaré oversimplified by suggesting that it is mainly a matter of submitting definitions, in such a way as to keep theory as simple as possible.

But their simplifications, even if going too far, honor the insight that the parameter that is measured is identified in the historical process by the *envisioned eventual* stable measuring practice, while it is differently identified *in retrospect* by the theory that draws on that history for its credentials.[44]

Within the vantage point of the accepted theory, once such stability has been achieved, we can speak meaningfully of the accuracy with which a given instrument gauges a given observable. At that late point, when the parameter has found its place in the theory that has emerged now, it is of course a characteristic which is no longer defined by measurement, but by its role in the theory. That is right and legitimate also from an empiricist point of view, for this discourse draws on how objects, phenomena, and instruments are classified by (or classified relative to) the accepted theory.[45] It does not presuppose an impossible god-like view in which nature and theory and measurement practice are all accessed independently of each other and compared to see how they are related 'in reality'. The two ways of looking at the matter we must combine in a synoptic vision. The first is 'from within' the historical process in which measurement procedures and theory are stabilized. The second is 'from above' with the theoretical description of the domain including the measurement interactions already in hand, a stage achieved in that historical process but no longer involving any explicit reference to its own history.[46]

We are not done with measurement, even with this realization. There is still a whole battery of questions about measurement that are not answered by our escape from the simplistic notion of number-assigning. On the one hand, a measurement procedure is a physical interaction. On the other hand, the measurement outcome 'locates' the object or process measured in a certain space of possible states. The physical and intentional aspects of measurements are to be looked into separately, before we bring them back into a synoptic view of how theory relates to experiment and observation.

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6

Measurement as Representation: 1. The Physical Correlate

We have looked at measurement 'from within', and will now look 'from above': that is, try for a view that we can have after the measurement procedures, concepts, and theories have stabilized. This resting point in the conceptual and scientific development is, to be sure, fleeting and momentary only—yet marking a context in which some things can be taken for granted pro tem, where it makes sense to ask about the world-picture of currently accepted physical theory.

For the question *What is measured?* the most direct answer is *Physical magnitudes that characterize the objects measured.* At the point where the pertinent theory is stably established, what the relevant physical magnitudes are and how they are measured is itself specified theoretically. We cannot separate the questions *What is measured?* and *What is a measurement?*, and both of them have, eventually, specific answers from within the pertinent theory, which will classify certain interactions as measurements and their final stages as outcomes. The interaction in question I will call the *physical correlate* of the measurement. The criterion for what sorts of interactions can be measurements will be, roughly speaking, that the outcome must represent the target in a certain fashion—, selectively resembling it at a certain level of abstraction, according to the theory—*it is a representation criterion.*

Physical conditions of possibility for measurement

Recall for a moment Nelson Goodman's example of the Duke of Wellington's portrait that resembled the Duke's brother more than the Duke. It shows that what the painting is a painting *of* is not settled by resemblance.

Granting that, let's ask though how we should classify this example, while thinking of how Dürer's *Art of Measuring* depicts the perspectival painter's art. Perhaps the portrait resembled the brother more, because the brother was in fact the artist's model while painting the portrait? That would not have been so remarkable—perhaps the artist had noticed how much the brothers did resemble each other, and the Duke was not always available for sitting. Not only not remarkable: in no way threatening the status of the painting as the Duke's portrait. To take a more extreme case, Dante Gabriel Rossetti's painting *The Girlhood of Mary Virgin* depicts the Blessed Virgin in the house of her parents. But the model for Mary was Christina Rossetti, while Mrs. Gabriele Rossetti was the model for St. Anne. That does not make the painting a family portrait of the Rossettis. It is without a single doubt a picture of the Virgin Mary's childhood, although the painted figures—equally undoubtedly—resembled Christina and Frances Mary Rossetti more closely than either the Blessed Virgin or her mother.

What if the artist commissioned by the Duke of Wellington had not only used the Duke's brother as model, but had used one of Dürer's machines for drawing, so as to produce the painting? In that case we say that the drawing is a measurement outcome, but the measurement was surely not a measurement of the Duke! So now the "of" in the two contexts diverge: we will still grant the result the status of portrait of the Duke, however it was made, but not that the operation carried out was a measurement of the Duke.[1]

While the criteria for what a portrait is a portrait of are various, cultural context-dependent, and sensitive to social factors, that is not quite so for measurement, however similar they may be in other respects. In the case of measurement the physical conditions of the object-instrument interaction have more weight, and are in fact the subject for a good deal of foundational research in the sciences themselves.

Focusing on the physical correlate

The answer *What is measured are physical magnitudes that characterize the objects measured* can be elaborated, after science has reached the requisite stability, in the form:

> What is measured is a physical system of a certain sort. Being of a certain sort, this system is characterized by certain parameters, and a

> measurement made upon this system is a measurement of one of those parameters that characterize it.

For at that stable point in the joint evolution of theory and practice there is a theoretical *classification* in place, of the objects that can be measured, before those measurements are made. Even the humble substances to be found on my breakfast table today are already so classified, by currently accepted theory, that it makes sense to ask of any one of them whether it is a sodium compound or whether it contains electrolytes; and there are measurement outcomes that would answer the question.

While Reichenbach focused on the physical correlate of such theoretical notions as *geodesic*, we can now raise questions about the physical correlate of measurement *per se*. Measurement is an operation by whose means we gather information; but this is of course done by way of a physical interaction between apparatus and object, to play the information-providing role. Measurement always involves a physical interaction between 'object' and 'apparatus'; that interaction we can call the *physical correlate of the measurement*.[2] This distinction, between the measurement and the physical interaction by whose means it is accomplished, comes into force when we ask just what the theory, speaking in its own terms, must provide by way of a representation of the measuring process. If that interaction is in the theory's domain, the theoretical description will be of this interaction in the same terms as any other physical interaction, and involve no terms that signify anything intensional or intentional.

At some point below we shall narrow our focus to the measurement outcome, which is the information provided by the final physical state of the apparatus in that interaction. There the same distinction needs to be made: the final physical state of the apparatus is the *physical correlate of the measurement outcome*. That is not a matter of meaning, but concerns a physical precondition for meaning. No information is gathered at all unless the outcome means something; but it can't mean anything relevant unless it appears at the end of a suitably structured physical interaction, and takes a particular sort of physical shape.

Let us concentrate on the case in which physical theory is taken broadly enough so that this interaction is in its proper domain. We take it for granted then that the theory provides a representation of that physical interaction, in physical terms.

We can put this partly in terms of language. A claim of the form "This is an X-measurement of quantity M pertaining to S" makes sense only in a context where the object measured is already classified as a system characterized by quantity M. To so describe an object is already to classify by theory. Therefore the claim is theoretical or at least theory-laden, and has to be treated as such.[3] It is a claim which will change in content and in truth conditions as our accepted theories change, since our classification of physical systems changes along with the theories. Here too we see a difference in the views from within and from above. From the *posterior* point of view the new theoretical claims are not displacements but refinements of the old—precisely because the old claims are now seen as (usually in some way imperfect) versions of less precise claims formulable in the new theory.[4]

This context-dependence of the claim may not be visible when we discuss a measurement procedure that was stabilized so long ago that we now take the description of the relevant objects and apparatus for granted. No one today would think of the use of a thermometer as being a meaningful practice only within certain theory-laden contexts. The practical use of these instruments has attained such stability that it is *for now at least* not at all sensitive to changes in the background theories. But that context-dependence springs to the eye in cases where the relevant theory is relatively new and has some claims to descriptive completeness—such as relativity theory or quantum mechanics.[5] In both of these, the behavior of the measurement apparatus is indeed within the domain of the theory itself, and the criteria for qualifying as [physical correlate of] measurement are to be presented in the terms of the theory—with consequent *questions of coherence and consistency* inevitably forced on us.[6]

Philosophical reactions to this point have included a good deal of extremism. There has been outright denial, as in the operationalist and early positivist delusion that we could have a hygienic 'observation language' in which the measurement operations and their outcomes could be described free from all theoretical content. Though we still sometimes see the logical positivist and logical empiricist circles of Vienna and Berlin identified with this point of view, it was in fact quickly abandoned there. The critique by Hanson, Toulmin, and Kuhn in the fifties and sixties removed it from the philosophical scene altogether—or should have.

The opposite extreme in denial came then, ostensibly in a famous chapter by Kuhn: scientists in different centuries live in different worlds.

That extreme is still worth some scrutiny even now. Understood in its most blatant literal sense, it involves a denial that there is any such thing as the physical interaction prior to or independent of the theoretical context in which it is classified as such or such a measurement. Just to say this shows it up for what it is; but for that very reason, shows that this was not what was meant. The real point is that occurrence of the physical interaction is one thing, its playing the role of representation of the value of a certain parameter is another; and today everyone understands that.[7]

To escape from both extremes, we need to separate quite clearly the historical process, examined in the preceding chapter, in which measurement and theory are jointly stabilized, from the retrospective characterization of measurement in the eventual established physical theory.

The question of coherence

How is the measurement interaction represented in the theory? The answer will be different depending on which measurement interactions are meant, and which theory. Stepping back from the specific cases for now, we ask the more reflective question: *what must the answer be like, if it is to be satisfactory?* The main criterion must surely be this:

> the theoretical characterization of the measurement situations is required to be coherent with the claims about the existence of measurement outcomes, their relation to what is measured, and their function as sources of information.

For this coherence it is required that the theoretical characterization of the measurement interaction allows for a coherent story about how its outcomes provide information about what is being measured. These coherence conditions we need to spell out when we characterize the physical correlate of measurement, in a context where it is assumed that both sides of the interaction have already been well understood and classified.

Recall that "This is an X-measurement of quantity M pertaining to S" makes sense only when the object measured is already classified as a system characterized by quantity M. This will be so if the context of discourse is governed by a theory; and it is in this kind of context that we can sense a challenge to consistency or coherence. What does this theory, or an accepted background theory, imply about the measuring process whence the theory reaps its credentials?

On the simplest view, the outcome of a measurement of quantity M on system S *reveals* the value that M has at the time the measurement is made. Even in the most familiar cases, however, this must be qualified. The process takes time, so is what is revealed: the value of M at the outset, or at the end, or . . . ? The final reading on a thermometer placed in a cup of tea we can take to correspond to the tea's final temperature. Why? Because the theoretical description of the interaction implies, via a pertinent conservation principle, that the two end up with the same temperature. The coherence of this reading of the thermometer is thus underwritten by theory. But the question that was most likely being asked was how hot the tea *was*? In other quite familiar cases, measurements are destructive: the patient does not survive the test [unaltered], the photon is absorbed, the metal sample fractures or is vaporized. Then the measurement does not reveal the final state of the object measured, since the object exists no more. But the test result should at least show whether, for example, the patient did have one illness or another.

Measurement can interfere or destroy—and perhaps the most direct indication it typically yields is of the state of the object at the end. But such a process will not serve its function as *measurement* (rather than, say, as *preparation*) unless some information can be gleaned about what the target was like before the measurement—isn't it?[8] So, granting that the outcome has its significance as such only in a theoretical context, let us ask: *what information can be derived* from the measurement outcome, concerning the initial state of the object, *sub specie the theory*? In the case of the thermometer the question would be: given the final reading, what can be inferred *via thermodynamics* about the tea's initial thermal state?

It looks like circularity threatens at once: to infer the initial temperature of the tea, we'd have to know the initial temperature of the thermometer, among other things. But the circle is easily broken: thermometry already relies on some identified fixed point, such as the freezing point of water, and the thermometer can be prepared to be initially in thermal equilibrium with such a fixed reference point. Can the circularity be so easily evaded in general? I think not. The familiarity of these classical examples and the stability long since achieved in their treatment may tempt us to assume that, but in fact, there are puzzles to be faced.

General theory of measurement

The greatest felt tension between the measurement procedures that we know by acquaintance and their description in a theory is found in quantum mechanics. Not surprisingly, therefore, that is where we find also the most extensive treatment of the physical side of measurement. Under the heading of "quantum theory of measurement" we do not find solutions to philosophical problems.[9] Instead we find there simplified and idealized but straightforward accounts of the physical [part of the] process. These accounts have a general form that has considerable claim to universal applicability in the empirical sciences. I shall take that form, depicted non-technically, as my central example of theoretical descriptions of measurement.

Initial measurement set-up

A proper measurement process must involve, besides the object on which the measurement is to be performed, a separate system to play the role of measurement apparatus. The procedure is meant to address a particular *parameter*, *property*, *quantity*, or *observable* (essentially interchangeable terms) that pertains to the object. The apparatus is to be coupled to the object in such a way that a process will occur that counts as a measurement.

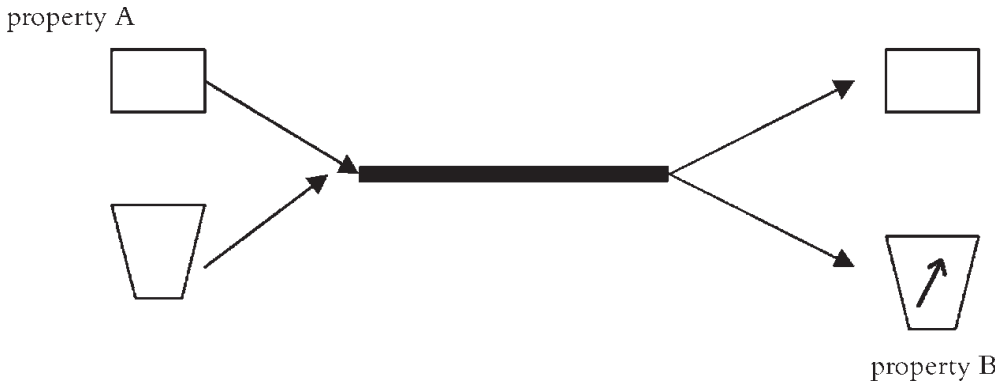

Figure 6.1. Measurement Schema

What are the criteria? If that process is a measurement of property A, say, then there must be a corresponding property B, of the apparatus,

whose final value will be the outcome. That process has to be governed by an equation which has to guarantee something about properties A and B and that final value—but classical intuitions beware! In addition, the system denoted as apparatus has to be such that it will play this role quite independently of what state the object is in—give it any object in any state and it will play its role.[10]

What must the governing equation guarantee, in order that this process really be the measurement of some property pertaining to the object? In a classical context we might say something like: if the object is thus or so then the end-state of the apparatus has to indicate precisely that.[11] This assumes that the parameter measured already has a definite value before the measurement, a value that can be revealed by the measurement outcome. But this "revelation" assumption is at best controversial and certainly generally rejected in foundational studies of quantum mechanics.

Measurement does not 'reveal' (?)

Complacency about measurement was sorely tried as the quantum theory developed. There are now many arguments in the literature to show how that theory requires a revision in our demands for what constitutes measurement. Instead of repeating these here, I will illustrate such reasoning with a simplified example that displays the sort of cautions we must entertain.[12]

Consider a strange situation characterized by three physical parameters, A, B, and C. The data are that, for each, measurement yields values 0 and 1 with equal frequency. We are able to measure them two at a time, and always find opposite values. Here is a simple theory that generalizes on these data:

> For X, Y = A, B, C: the probability that X has value 1 equals 0.5, as does the probability that it has value 0. But the probability that X and Y both have value 1 equals 0 if $X \neq Y$

But this simple theory is self-contradictory. For it assigns probability 1 to both of the following:

> No two of A, B, C have the same value
>
> Each of A, B, C has value 0 or 1

It is impossible for both to be true! What went wrong? The data listed were about measurement outcomes, while the extrapolated probabilities were absolute and unconditional. The proper generalization must give due attention to the fact that what was found was how things appear *in the measurement set-up*:

> For X, Y = A, B, C: the probability that X has value 1, *conditional* on its being measured equals 0.5, as does the probability that it has value 0. But the probability that X and Y both have value 1, on the *condition* that the two are measured together, equals 0 if $X \neq Y$

Now there is no contradiction, but rather the consequence that A, B, C cannot all be measured together. So that is our solution.

How good a solution is it? We want to ask, surely, what value C has while A and B have values 1 and 0 respectively. But how this question is to be understood depends on what we take for granted here. Can we take for granted that when not measured, A, B, C still can only have values 1 or 0? Or could they have other values when not measured? Or perhaps have no value at all? And secondly, can we take for granted that if a measurement of A shows value x, then at that moment, A has value x? Or that it had value x at the beginning of the measurement?

Could we perhaps reason like this? Suppose A is measured and the outcome is 1. Now we can predict that a measurement of one of the others, B or C, will have outcome 0, with certainty. On this basis, can we assert that B and C already have value 0 at this moment? If we do, we will have to add that joint measurements of B and C that are actually carried out are *systematically deceptive*, for they never show them having the same value.

Before seeing an example like this, or at least before having any inclination to take it seriously, various assumptions involved in such reasoning would likely have been taken for granted when thinking about measurement. Classical intuitions (if such beings exist) suggest two postulates:

Value Definiteness: Each physical parameter always has some value, namely one of the values which may be found by measurement.

Veracity in Measurement: Measurement of a parameter faithfully reveals the value it really has.[13]

These two postulates can be consistently added to our above story, but then they imply that either measurement is subject to or involves a systematic alteration, or else some sort of conspiracy in nature constrains when measurements are made: when A and B both have value 1, we are lucky or clever enough not to measure the two of them!

Perhaps we should put it this way. The conjunction of these postulates would be an attempt to say that the world is basically the same, whether things are being measured or not. But given the above story, the two postulates are both true only if things are not basically the same whether things are being measured or not! So the attempt fails: some difference between measured and unmeasured world will have to be admitted as a possibility here.

Paul Feyerabend's (1958) name for the postulate of *Value Definiteness* was *Classical Principle C*. This principle must be rejected, he argued. From Bohr to Feynman, physicists have expressed similar opinions: an observable (measurable parameter) might not have a specific value outside the context of measurement. However, the second postulate—*Veracity in Measurement*—has also been much looked upon as a candidate for rejection or revision. To keep the first postulate and reject the second, by means of an explanation through uncontrollable disturbance by measurement, would not be a happy option. It would imply some sort of conspiracy again: if A and B do sometimes both have value 1, how does the "uncontrollable" disturbance in measurement carefully and systematically hide that fact?

Finally, rejecting *Value Definiteness* would by itself already imply a weakening of *Veracity in Measurement*. For if, at a certain time, parameter A has no value, and is measured, then this measurement *yields* a value as outcome, but clearly does not *reveal* a value.

When we specify what counts as a measurement of A, we describe a physical arrangement which must have one of two outcomes (indicator values), in this case 0 or 1. For this to be a measurement, hence to play a role in information gathering, it must surely do something that is revealing about what is measured? But what sort of information it does yield, and how much, we shall have to consider very carefully.

General theory form

What the apparatus is like at the end must reflect some pertinent feature of what the object was like initially. There must be a transfer of some character

of the object's initial state to the apparatus' final state.[14] Otherwise there would be no way to use this process to gather information about the object on which the measurement is performed. How this 'transfer' requirement is to be made precise will of course be different in different theories.

For the general form we must allow for the case in which the relation between physical state and measurement outcome is only characterizable in terms of probabilities. A deterministic theory can be thought of as the special case in which all probabilities are zero or one. The measurement situation modeled as theory prescribes, when the apparatus is itself in the theory's domain of application, must include a specification of the following factors:[15]

> a family M of *observables* (physical magnitudes) each with a range of possible *values*;
>
> a set S of *states*—physical states of both the system measured and of the measuring system;
>
> a *stochastic response function* P_s^m for each m in M and s in S, which is a probability measure on the range of m; with P_s^m to be interpreted as *the probability that a measurement of m will yield a value in E, if performed when the state is s*.

Suppose now that one sort of process represented in the theory is that of the interaction that qualifies as measurement of an observable. The situation depicted then involves two systems, the object measured S and the apparatus R by which it is measured. Together S and R constitute a larger system, a 'two-body' system, S + R. The family of observables must then include some that pertain just to that object S, some that pertain just to the apparatus R, and some that pertain to both at once, that is, to system S + R—and similarly for the states. But there must also be a constraint on how this situation evolves, as the two objects are coupled and interact.

Classifying R as an apparatus, for measuring observable A for example (an observable pertaining to the object to be measured) entails that this interaction will take a certain form, which qualifies the designation as "apparatus for measuring A". In fact, this interaction must be such that *something pertinent about the initial state of the object is reflected in the final state of the apparatus*. So imagine the apparatus as having a dial with a pointer—as a system it is characterized in part by an observable B, the *pointer observable*,

whose possible values are precisely the numbers which the pointer can indicate on that dial. Now the criterial condition, in its strictest form, must be this:

> Criterion for the Physical Correlate of Measurement: $P^{B}_{fin}(E) = P^{A}_{init}(E)$

where *fin* is the final state of the apparatus, and *init* is the initial state of the object on which A was being measured. That is, the final state of the apparatus must reflect, in its probabilities pertaining to pointer observable B, the probabilities pertaining to measured observable A in the initial state of the measured object.[16]

This includes as a special case that the pointer observable B on the apparatus would most certainly show e.g. value 17 if inspected at the end, on the supposition that the measured object was initially in a state in which observable A 'most definitely' had value 17. To this extent then the old criterion of Veracity (or *revelation*) is being honored still.

The coherence constraint

But notice the qualifying clause "if inspected" and the qualifying quotation marks that we found it necessary to insert in the preceding paragraph. For the probabilities $P^{B}_{fin}(E)$ and $P^{A}_{init}(E)$ are understood as conditional probabilities—conditional on measurement!

At this point one begins to have the sense that there may be ways into the enchanted wood but no ways out. What does the Criterion for the Physical Correlate of Measurement come to when we look at it in this light?

Firstly, the theory will have implications for results of other measurements (of observables A′, A″, etc.) on systems with the same sort of preceding history, depending on the results from the A-measurements. This is one way in which inconsistencies could arise with the theory when different measurements are made, so we have here an empirical coherence condition. This condition depends so much, however, on the specifics of the theory in question that it may not be possible to say more about it in general.

Secondly, the theory must imply that if *two other apparatus* were coupled to these systems, to the object at the beginning and to the apparatus at the end, the probabilities of the corresponding outcomes would be the same.

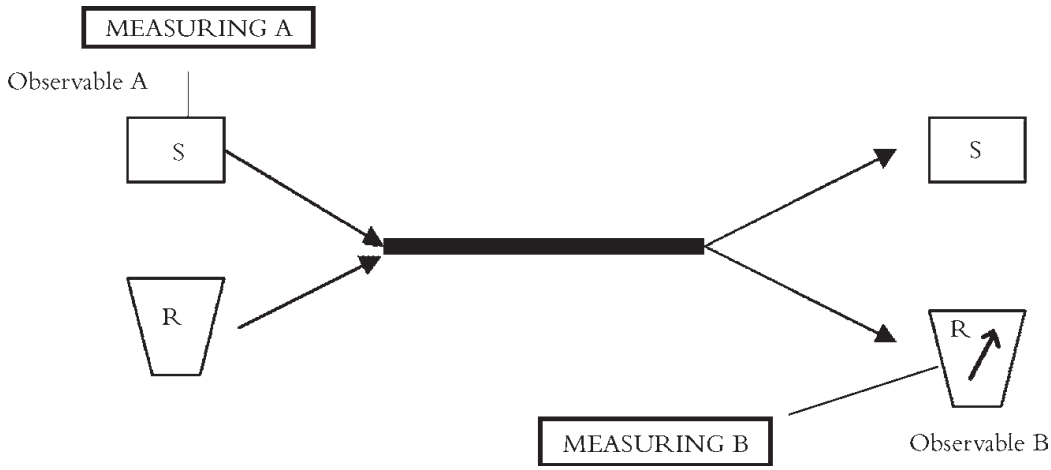

Figure 6.2. Coherence of Measurement

That is, suppose that we had two further apparatus, R1 and R2, the former one also used to measure A on system S and the latter to measure B on apparatus R. Then the Criterion's being satisfied *mutatis mutandis* for all measurement interactions will guarantee the coherence of the 'exterior' measurements with the 'interior measurements'.

For a comparison of the probabilities pertaining to R1 and R2 should show an accord that 'reveals' that the accord between S and R demanded by our Criterion was satisfied. The theory must predict that if a second independent measurement was made to secure the premise that the object had been prepared in a certain initial state, and a third measurement were made of observable B on the apparatus end-state, then the demanded accord would be found with the requisite probabilities.

This is therefore a **coherence constraint** on the theory: it must first of all have this internal harmony in what it predicts, but secondly, comparisons of results in the two kinds of set-ups must be empirically vindicated. So here we have a coherence condition that is partly internal consistency and partly empirical. The theory is to satisfy that coherence constraint—and that is the most we can ask of it. Put in terms of the sort of description the theory provides for physical systems in general, this 'physical theory of measurement' is not plagued by any sort of circularity.[17] At the same time, of course, it does not pretend to aim at a definition of measurement in terms that have nothing to do with measurement!

In practice, the level at which a theory confronts experience is not that of raw data taken from individual measurement outcomes, but of the 'data

models' constructed on their basis, and the further *smoothing* of the data models in which for example sequences or discrete variables are replaced by continuous functions.[18] But the conceptual problems—such as the 'measurement problem' of quantum mechanics—refers to the individual outcomes of measurement interactions, modeled in the above fashion. I will return to this distinction between data and data models below. The point is that, however we conceive of this, the above coherence constraint will have to apply to how the data from measurement are to be accommodated by the theory's theoretical models.

Veracity reconsidered

The criterion imposed on the physical correlate of measurement is as strong as possible, given the general form of a physical theory that was under consideration. That form, in turn, was kept very permissive, so as to allow for the form that quantum mechanics took when it was formulated definitively *c.* 1925. As a result, the conditions on measurement had to allow violations of the two 'classical' principles that we had noted: *Value Definiteness* and *Veracity in Measurement*.

As to Value Definiteness, nothing in the Criterion for the physical correlate of measurement precludes the observable A to have no value (or only an 'unsharp' or 'fuzzy' value) at the outset. In fact it is not even implied that the pointer observable B will have a specific one of its possible values in the final apparatus state. All that is implied is that if measurements be made, to measure those observables, the possible values will appear as outcomes with certain probabilities. Value definiteness is not implied in any sense, way, or form.[19] Veracity is implied only in the very much weakened form of accord among conditional probabilities. But if the theory specifies nothing beyond those conditional probabilities then no stronger criterion can even be formulated for the physical correlate of measurement, to the extent that the theory can cover that.

If we now return to the empirical assertions, we are not bound to stay within the theoretical description, and we can refer freely to the actual outcomes of measurements, as typically summarized in data models. These data models are constructed from the raw data that are actually gathered, so we are dealing here with actual frequencies, and probabilities that have a good fit to those frequencies. Suppose now that such a 'summary' is pretty well a picture of a *state* in a theoretical model, to the extent of

displaying probabilities derivable from that state. Then the theory may, and generally does, have implications for how that state evolves in time.[20] Thus predictions can be made about what will be found if new measurements are made in that situation at a later time.

As an example, we can take the Stern–Gerlach apparatus, named after Otto Stern and Walther Gerlach's famous 1920 experiment. There is a classical picture behind the idea of the experiment: imagine a 'beam' of particles being emitted, with a particle being like a classical dipole with two halves of charge spinning quickly. In a magnetic field, such a particle will begin to precess. This way the particle's position becomes perfectly correlated with its spin value. That is also the result, *mutatis mutandis*, with the situation described in quantum mechanics. When the spin in a given direction can have only two values, the beam is split into two separate beams, 'upper' and 'lower'. The early attempts to realize the experiment encountered many difficulties (cf. Bretislav and Herschbach 2003). In the first realization a beam of silver atoms (produced in an oven, at temperature 1000 °C) was collimated by two narrow slits (0.03 mm wide) and traversed a deflecting magnet 3.5 cm long with field strength about 0.1 tesla and gradient 10 tesla/cm. The splitting achieved was only 0.2 mm, and there were doubts as to the data obtained. (Of course the set-up is now described in rather more ideal terms when it is used to illustrate quantum properties.) The apparatus can be rotated, so as to measure spin in any direction. Thus data on different spin observables can be collected on some samples produced in the oven, and on the basis of the frequencies in those samples, it is possible to infer—via the theoretical description and classification of this process—just what state is prepared by that source.[21] Then the proportions of the output in the two channels in later measurements can be predicted on that basis.

It would be illegitimate to conclude that the silver atoms exiting in the upper channel were prepared in the oven in a state of spin-up. Rather, the oven prepares a beam of atoms in a state which is such that the probabilities of a Stern–Gerlach measurement having outcome *up* or outcome *down* are definite, with the result that the relative proportions in the two channels are definite. But still we can see now that *Veracity* is honored at some appropriate level. The outcome does not reveal a prior state for an individual silver atom, but the frequencies in the outcome do give information about the prior state in which the source prepares what it

sends out. If it were not so, the role of measurement would not be played at all, since the outcomes would not serve to provide information about what the measurement is performed on. In practical terms it is precisely the source on which the measurement, taken as a whole, is performed.

What is not measurement

In discussing the physical correlate of measurement we are displaying physical preconditions, conditions for the possibility of measurement. These do not define what measurement is, and in fact there will always be many interactions conceivable in nature that satisfy those conditions but are not measurements. For measurement is information gathering, a measurement outcome is something that has meaning, is in fact a representation of what is measured, and that point does not reduce to a physical condition.

In the foundational literature on physics, this distinction is mostly honored by neglect—reasonably so, since the focus there is on what is within the intended domain of application of physics. In the sense of "measurement", as the term is typically used there, there are (according to the pertinent theory) measurements going on all the time in the stratosphere and at the sub-atomic level, far away from what we can do or use. So in this usage of the term, there is no reference to practice or to our information gathering.

On the macroscopic level too we can think of processes that connect two situations separated in time or space. These could be so correlated that it would be possible in principle to get information about the one by inspecting the other—provided of course we knew of that correlation! But that something *could be done* does not mean that *it is done*. That something could be assigned a representational role does not mean that it has one. There are contexts where the distinction between a measurement and its physical correlate can be neglected, but philosophical reflection is not one.

7

Measurement as Representation: 2. Information

A measurement outcome is something physical: an event, the end-state of the apparatus, or an object (photo, graph, list of numbers) produced by the measurement process. On the other hand, measurement is information-gathering, so a measurement outcome has a meaning (i.e. information content). The information provided is "read" off the measurement outcome, which is a physical state or event, but one with meaning to the literate.[1]

The term "measurement" is an endorsing term. If we call something a measurement, we imply that there is something correct or valuable in the way it yields a representation (even if on this particular occasion it went wrong). Measuring is an operation by which we can produce or gather information; and here "information" too has an endorsing sense. The communication engineer's neutral usage of "information" has begun to modify common usage to some extent. But even now it would still be puzzling or provocative to hear "We get information from observation, measurement, fictions, lies, and popular astrology." If we insist that measurement is information gathering, we mean in part that fictionalizing or speculating or guessing is not measurement.

Measurement is an operation, using something that functions as an instrument, to gather information. The instrument is being used, in Heidelberger's terminology noted above, in a representative role, that is, the operation has as outcome a representation of the object or situation operated on.[2] Since many sorts of operations can have such outcomes, this notion needs to be narrowed. Attention to more complex measurement situations, that need to be assimilated via a series of steps in model building, leads—I shall argue—to the need for a broad concept in which measurement is itself a specific form of [self-]location.

What is measurement—number-assigning?

We do find simplistic answers sometimes even in places where it matters.[3] Lord Kelvin, with a well-deserved reputation as scourge of purported sciences outside the ken of physics, wrote:

> I often say that when you can measure what you are speaking about and can express it in numbers you know something about it; but when you cannot measure it, when you cannot express it in numbers, your knowledge is of a meager and unsatisfactory kind [and] you have scarcely . . . advanced to the stage of science . . . . (Thomson, Lord Kelvin, 1891: 80–1)

This sort of view was developed more systematically by Norman Campbell (Campbell 1920, 1928, 1943) in his influential writings on measurement. More historically important to the subject is S. S. Stevens's seminal paper for the behavioral sciences in *Science 1946,* which appears at first sight to express Kelvin's view, for he writes

> [W]e may say that measurement, in the broadest sense, is defined as the assignment of numerals to objects or events according to rules. (Stevens 1946: 677)
>
> [W]e may venture to suggest by way of conclusion that the most liberal and useful definition of measurement is . . . 'the assignment of numerals to things so as to represent facts and conventions about them'. (Ibid., 680)

Stevens's article was responding to a commission which had attempted to arrive at a general, comprehensive account of measurement. The focus in that historically important paper was quite narrow: the mathematical foundations of measurement, in the sense of a study of *conditions under which an algebraic structure is embeddable in the real number continuum*.[4] Almost all of his discussion pertains solely to measurement scales and the sort of invariance that is required for meaningful scaling, and is only superficially linked to either the process or the role of measuring in scientific practice. The algebraic structures in question are described informally in terms of physical *addition* operations, but this is kept skimpy and abstract. Stevens is able to maintain those slogan formulations only by, on the one hand, extending the concept of number beyond its normal use, and on the other hand, by restricting himself to the abstraction involved in a mathematical point of view.

In fact the idea of an essential link between measurement and number-assigning comes from taking a myopic view of mathematics and an equally myopic view of measurement.

Mathematics is not the study of numbers. Numbers are merely a particular if useful and user-friendly instance of mathematical structure. Granted, the numbers' salience is not the only reason to think of measurement mainly in terms of numerical outcomes. Some paradigmatic examples of measurement are indeed numerical. But as Dürer's *Treatise on Measuring* already brought out, we will have to look beyond those.

Kinds of scales

The guiding idea for the study of measurement scales is that the grading must be thought of as reflecting characteristics of operations on something physical—operations that can plausibly be called measurements in some 'ordinary' sense. Presumably Stevens was referring to this with "according to rules", though what sort of rules for operating on real physical things count as measurement recipes was left aside. The constraints on the possible outcomes that he describes are designed so as to guarantee that what can be measured is precisely what can be represented as graded on a real-number scale.

Mathematically speaking there are many structures other than the real number continuum. Even in the field called "foundations of measurement theory", the conception is already so generalized that the standard numerical version can only be seen as a particular case. The authors in that field are aware of the limits of numerical representation, and Stevens himself introduced the now commonly found distinctions:

nominal measurement is the assignment of (numerical) labels without implying any algebraic structure;

ordinal measurement assigns a rank ordering;

interval measurement is ranking on a scale where only the intervals between elements are numerically comparable;

ratio measurement is ranking on such a scale where there is also a minimum, and the ranking can be represented by non-negative numbers with the ratios between these numbers reflecting a physical relationship as well.[5]

Each of these categories has its examples. The Mohs hardness scale for minerals is the typical example of an ordinal scale.[6] If we ignore absolute temperature then our thermometers, whether Fahrenheit or Celsius, provide an interval scale—the Kelvin scale for temperature with its absolute zero is a ratio scale.[7]

But Stevens's taxonomy is not exactly a table of categories supported by a transcendental deduction! It looks nicely hierarchical: we can suppose that ordinal measurement must be a special case of nominal measurement with the labels reflecting the ordering. Then each category besides the nominal presumes an ordering which is linear: two assigned labels x and y must either be the same or have one greater than the other.

As was much emphasized in nineteenth-century discussions, notably by Mach and Poincaré even this ordering requires a contingent empirical regularity for its coherence.[8] To be able to order at all, even in its most minimal logical sense, one needs at least a criterion of equality. Suppose this be specified, and suppose that by this criterion A and B are both equal to C. Does it follow that by the same criterion A will be equal to B? No, it does not, at least not logically, if the criterion refers to a performable physical test. But the relationship of equality, that is of having the same position in the ordering, must be transitive or the ordering falls apart. (Similarly of course for the relationship of *greater than* or *after* if applicable). What criterion is proposed is up to the proposer, and if there is a a variety of plausible candidates for the criterion then this may be a matter of choice or even convention. But whether a proposed criterion can be adopted will depend in part on contingent empirical regularities, on pain of incoherence in practice.

Returning now to ordinal measurement: the numerical relationships between the assigned numbers are not all of them significant. If we rank-order some items by assigning the numbers 1 to 10, merely on the basis of some 'greater than' relationship, there is no significance implied by the fact that 7 is as much greater than 5 as 5 is than 3, or that 6 is 3 times as great as 2.

What is significant is what is invariant under 're-scaling'. This is not an explanation; it is a remark that connects various notions connected with each other. Take the example of a nominal scale: the members of a soccer team are numbered 1 through 11. It is important only that different players receive different numbers, the ordering does not matter. That is equivalent

to saying that all permutations of this assignment result in equally admissible numberings: each such transformation is an admissible re-scaling. But what decides what is and is not important, what are precisely the relevant features to be preserved in all and only the admissible transformations of the scale, is not decided by mathematical characteristics. It is decided by two factors: the measurer's purpose and the relationships that can be accessed instrumentally or practically on which to base the assignment. These factors are outside the domain of the mathematical theory of measurement.

Significance and invariance

There is an important difference between what we might call "scale invariance" and "scaling invariance". In the preceding Part, in the discussion of scale models, we saw that a perfectly proportional decrease or increase in geometric shape may be quite the wrong thing for scale modeling, depending on what is meant to be studied or predicted. This is a point about "active" transformations, in which e.g. something literally of great size is replaced by something of small size for the purpose of some research. But *scale transformations* such as the permutation of a nominal assignment, or change from centimeters to inches in length measurement, are just changes in bookkeeping. The books must be kept straight—what is important must be properly represented; but this is a purpose-dependent norm for representation, not something telling about nature.

Significance is to be equated with invariance. That is, the procedure by which the numbers are assigned may favor only one large class of ways to assign them, as against other assignments, and only what is common to the favored class matters. This was an important insight exploited by Stevens: the types of scales are to be distinguished in terms of the families of transformations under which they are invariant.[9]

Interval measurement illustrates this well. An admissible transformation is in this case a combination of a dilation (in a sense that includes contraction) and a translation. As familiar example take the Fahrenheit (F) and Celsius (C) "centigrade" scales for temperature. Suppose that the high today is 41 °F and yesterday's was 32 °F. Does that mean that the temperature today is more than 20 per cent higher than it was yesterday? The scale transformation from F to C is this: $F = 32+(9/5)C$. So today the high was 5 °C, and yesterday it was 0 °C. The ratio of today's temperature to yesterday has no invariant sense. But in ratios of intervals we find an

invariant. So suppose that on the third day it is 50 degrees Fahrenheit, so we say that the increase was the same each day. Then in Celsius, carefully calculated, we find the same: the increase from 0 to 5 was followed by an increase from 5 to 10.

The rules for measuring temperature allow a large class of number assignments. In fact, the only constraint on this a class of numerical assignments is that it is closed under operations of translation and dilation. That means: if Y is an admissible scale then aY+b is too.

So what significance can a thermometer reading have? The answer, as we also mentioned, was that interval ratios are invariant:

$$(aY_2 + b - aY_1 - b) / (aY_4 + b - aY_3 - b) = (Y_2 - Y_1)/ (Y_4 - Y_3)$$

There is a way to express the significant part of what is found in a temperature measurement with single numbers. That is to choose one interval that can be stably fixed—e.g. the freezing and boiling point of water—as reference, and pay attention only to the ratio of the intervals in which that is divided by the temperature reading. If the numbers assigned to those fixed points are 0 and 100, then the reading will be numerically equal to the percentage selected of the referent. Thus the Celsius scale is the most convenient: a reading of 15 °C indicates saliently the proportion (the interval from 0 to the reading point)/(the reference interval) as 15 per cent.

The choice of fixed reference points is conventional, subject only to the possibility of stabilizing the measuring procedure to that extent—and thereby hangs a tale.

The connection between invariance and significance (often referred to, somewhat confusingly, as "meaningfulness") was studied at length in the literature on measurement. There is an important connection between this subject and the technique of dimensional analysis which we encountered in the discussion of scale models. The seminal article by Duncan Luce (1978) begins with the question why the laws in physics are required to be dimensionally invariant, that is, invariant under the admissible scale transformations. As focus he takes the case of measurement scales:

> It is pretty well agreed that the problem of meaningfulness within a single measured attribute is closely tied to knowing how two different, but equally acceptable, numerical representations of the qualitative structure are related. Meaningful

numerical statements are those that remain invariant under permissible changes in the representation. That, however, is very similar to saying, in the more complex case of physical laws, that the representation must be dimensionally invariant. So, on the face of it, there appears to be a close relationship between meaningful statements in a single attribute and dimensionally invariant laws stated in terms of several attributes. (Luce 1978: p. 3)

The result he proves is in effect that features of a relational structure are invariant under the symmetries of that structure precisely if they are representable by a dimensionally invariant numerical function.[10]

Approximative measurement

We need not go far afield to find that linear ordering in a scale is too restrictive, in general. After all, a measurement outcome is not infinitely precise: the length of the table is registered e.g. as 100 plus or minus 1 cm. So here the real outcome is not a number but an interval. These intervals are ordered by inclusion, which is a partial rather than linear ordering. We can introduce a notion of 'strictly greater', e.g. by the definition that one interval is *greater* than another if all its elements are greater than all of the other's elements. But of course it does not follow then that the ranks are the same if neither is strictly greater than the other—so the ordering is not linear.

If this practical point is granted, we have already left behind the idea that measurement outcomes can always be represented as points on a linear scale. For the class of regions—however delimited—that can be indicated as found locations in this way is not a linearly ordered class. It is a class partially ordered by set inclusion, or by set inclusion modulo differences of measure zero, or some such relation. We should not call this "locating on an interval scale"—that term already has an established use, as we saw above. Rather, in this case, the object is located in the space of intervals, or in the larger space of 'Borel sets' generated by countable meet and join operations. This is the range most typically encountered where measurement results are not assumed to be 'point-like sharp'. There an elementary form of statement, which can be either that of a theoretical assertion or of a measurement outcome, relates a physical parameter to a set of its possible values, which is itself linked to a defined region in a much larger state space specified by a theory.[11]

The scale as logical space

Consider a simple theory of gases which characterizes them in terms of three parameters, P(ressure), V(olume), and T(emperature). This theory provides a three-dimensional mathematical space, and measurement will locate given bodies of gas in regions of this space.

The first point to notice is that this use of measurement makes sense only in the context of the theory in question, which already provides the general framework for classifying the items to be located in that space. Secondly, this is an example in which the classification is very simple and particular. In general, measurement of an item classified as being in the domain of a particular theory will locate that item rather indefinitely as somewhere in a space common to a whole family of models provided by that theory. The farthest but still illuminating generalization of this point was made by Wittgenstein, inspired by his study of thermodynamics and mechanics:

1.13 The facts in logical space are the world.

2.013 Every thing is, as it were, in a space of possible atomic facts. I can think of this space as empty, but not of the thing without the space.

2.0131 A spatial object must lie in infinite space. (A point in space is an argument place.)

A speck in a visual field need not be red, but it must have a colour; it has, so to speak, a colour space round it. A tone must have *a* pitch, the object of the sense of touch *a* hardness, etc.

2.202 The picture represents a possible state of affairs in logical space.

The HSB color space, with dimensions hue, brightness, and saturation is a good example of a logical space, but so is the PVT space in elementary gas theory, phase space in classical mechanics, Hilbert space in quantum mechanics; space and time themselves also serve as examples.[12] I submit the following generalization as the proper concept of a measurement operation:

> measurement is an operation that locates an item (already classified as in the domain of a given theory) in a logical space (provided by the theory to represent a range of possible states or characteristics of such items).

The act of measurement is an act—performed in accordance with certain operational rules—of locating an item in a logical space.

While the terms vary, descriptions of actual measurement procedures typically bear this out. This has been better appreciated in philosophy of technology than elsewhere. Thus Davis Baird:

> Measurement presupposes representation, for measuring something locates it in an ordered space of possible measurement outcomes. (Baird 2004: 12).

When discussing the thermometer and, in more detail, the Indicator Diagram for a steam engine, Baird uses the phrase "field of possibilities": "we have on the thermometer's glass tube a scale that displays the field of possibilities that we embrace with the thermometer". (Ibid., 2003: 50) In the steam engine case, the indicator is an instrument that produces a simultaneous trace of the pressure and volume inside the working cylinder as the engine runs through its cycle:

> The instrument harnesses an instance of material agency, the behavior of pressure and volume in a steam engine cylinder as it goes through its cycle. [. . .] At the same time, the indicator presents information. A field of possibilities is constructed in terms of the pressure-volume graph on which the indicator "writes". (Ibid.: 52)

As a special case, the logical space can be a scale, which may indicate the location as being in a certain region of a larger space. Thus a pressure measurement locates a gas in a region of the larger PVT space, a momentum measurement locates a body in a region of its phase space. So the locating is typically not in an exact point, but in a region. We already saw this in connection with imprecise measurement, which assigns not numbers but intervals, hence takes its assignments from a partial rather than linear ordering.[13] But this holds more generally, since what is measured is usually only some aspect of that 'field of possibilities'.

Thus measurement is an act of locating an item in a logical space. The converse does not hold: you can locate me in the logical space of astrology simply by asserting that my Sun sign is Aries. Above I added "performed in accordance with certain operational rules", but by itself that only points to another question. The astrologer's or soothsayers's or visionary's operational rules may not count as yielding genuine measurement. What precisely is needed? That is precisely the heart of the 'problem of coordination', hence requires looking into the joint evolution of theory and measurement. But

we can add that once a stable theory has been achieved, the distinction between what is and is not genuine measurement will be answered *relative to that theory*.

Here is a good example: as Henry Margenau and Adolf Grunbaum discussed, there are certainly procedures that look like simultaneous position-velocity measurements of particles at any scale. But quantum theory classifies them as having no such significance—for no operational outcome can reveal characteristics that, according to the theory, the system cannot have (cf. Grunbaum 1957: 713–15).

Data models and surface models

It is typical of discussions of measurement in foundations of physics to retain the simplifying picture of a measurement as a single operation, a single interaction with the object to be measured. In actual practice, as pointed out above, that is quite unrealistic.

There are measurements that can be done in one shot: when the nurse takes the patient's blood pressure she does not make a series of measurements and then calculate the mean and standard deviation. But that is in part because no very great accuracy is required here, and in repeated results there is unlikely to be a spread of more than a few points. In the general case this tolerance of inaccuracy cannot be maintained, and a measurement will take the more complex form. The result obtained in one operation we may call a *datum*, the results obtained on one such occasion the *data*; these data are subjected to a statistical analysis to reach the *official* measurement outcome.[14]

Blood pressure, weight and mass are still very simple examples, where the outcome may be reported as a single number, or pair of numbers (e.g. mean and standard deviation). Other, still quite familiar, cases display greater complexity. On the weather forecast website I consult I can find a graph depicting yesterday's temperature plotted against time. This was constructed from data gathered at various stations in the region, at various times during the day—this graph is a smoothed-out summary of the information that emerged from all these data, it is a *data model*. The question about the daytime temperatures in this region of one day ago is answered with a *measurement outcome*, certainly—but that is the graph in question, which is a data model constructed from an analysis of the raw data.[15]

That this is how the results of measurement, and the complexity of their relation to theory, must be conceived was an early and continuing theme in Patrick Suppes' work:

> exact analysis of the relation between empirical theories and relevant data calls for a hierarchy of models of different logical type. Generally speaking, in pure mathematics the comparison of models involves comparison of two models of the same logical type, as in the assertion of representation theorems. A radically different situation often obtains in the comparison of theory and experiment. Theoretical notions are used in the theory which have no direct observable analogue in the experimental data. (Suppes 1962: 253)

Our words have a sometimes disturbing plasticity. At one of the local small stations a particular reading is made at a particular time: that is a measurement. But the total process of all those single data collections plus the statistical analysis plus the systematic 'summarizing' that finally yields the graph—that also is a measurement. It is the latter that takes a more scientifically significant form. What is important is that in both cases, the primitive single data collection and the complex complete measurement with a stable result, the outcome must be regarded this way: *this is what the object looks like in this measurement set-up*. And in both cases, the object measured is thereby located in a logical space, characteristically associated with the type of operation involved.

Suppes continued with "In addition, it is common for models of a theory to contain continuous functions or infinite sequences although the confirming data are highly discrete and finitistic in character." The graph or other summary of the data found will be abstracted into a mathematically idealized form before it reaches the theoretician's desk. Let's call that processed artifact a *surface model*. Although usage does not restrict the term "data model" to summary representations of frequencies, let us take that as our example. I distinguish *data model* and *surface model* in this way:

- the data model summarizes the relative frequencies found
- the surface model 'smoothes'—in fact 'idealizes'—this summary still further so as to replace the relative frequency counts by measures with a continuous range of values.

The term "data model" is often used in the more general sense that does not distinguish summarized relative frequencies from probability measures, and

I will not insist everywhere pedantically on this distinction. Probabilities relate to relative frequencies—a subject all by itself—but the frequencies alone are what can be found in nature.[16] Any procedure for replacing a finite set of data by a graph furnishes an example, and such procedures are common throughout the sciences, facilitated by easily available software. But the replacement is a step in a long process from hands-on or mechanical data gathering to the eventual mathematical structures that can confront the proffered theoretical models.[17]

The complexity of the journey from raw data to graphical representation is also well illustrated by Ronald Giere's examination of astronomical observation. The data gathered by optical, radio, and gamma ray detection in today's astronomical observatories are combined and transformed in a process that yields visual representations color of comets, stars, nebulae, and galaxies. Giere examines how astronomical data, gathered in the form of black and white photos, are processed to yield images in color of a nebula, through a process originally invented by James Clerk Maxwell. Each individual photo captures just one aspect, but attention to the filtering by which it is obtained allows it to contribute to the reproduction of color in the images that are the final product. As Giere emphasizes,

> the images presented . . . are conclusions. These images present a picture that is continuous, or at least very fine-grained. The actual data cannot be that fine-grained. The data are made up of individual events recorded in various detectors at different times and processed by various physical and computational means. The images are constructed using those data, but go beyond the data. (Giere 2006: 48)

Recall that just now we are looking at measurement 'from above', that is, at this construction guided by accepted theory, rather than 'from within'.

Surface models and their embeddings

A theory provides, in essence, a set of models. The "in essence" signals much that must be delicately expanded and qualified; I will leave this aside for the moment.[18] These models—the theoretical models—are provided in the first instance to fit observed and observable phenomena. Since the description of these phenomena is in practice already by means of models—the 'data models' or 'surface models', we can put the requirement as follows: the data or surface models must ideally be isomorphically embeddable in theoretical models.[19]

Since the advent of quantum theory much thought has been given to the form that any possible surface model must take. Consider an experimental situation of a quite simple structure, involving several alternative measuring arrangements, a classification of possible outcomes, and some probabilities extrapolated from (imagined) observed frequencies. Then a surface model can be thought of as specified by three factors:

(i) two sets of observable conditions:
  (a) a set of realizable measurement choices—call it *PRC*, and
  (b) a set of possible outcomes—call it *PRS*;
(ii) the *surface state P*; this is a function which assigns probabilities of outcomes in *PRS,* conditional on measurements in *PRC*.

So P is defined on at least part of $PRC \times PRS$ and its values are real numbers in the interval [0, 1]. This structure is subject to certain minimal conditions which must guarantee that P is mathematically extendible to a classical probability function.[20] The numbers assigned by the surface state we can call *surface probabilities*.

What about theoretical models? We already looked at this briefly above. These need to be conceived without prejudice in favor of determinism or causal modeling. The theoretical model could specify, in general,

> a family M of *observables* (physical magnitudes) each with a range of possible *values*;
>
> a set S of *states*;
>
> and a *stochastic response function* P_s^m for each m in M and s in S, which is a probability measure on the range of m.

The number P_s^m is to be interpreted as the model's specification of the probability that a measurement of m will yield a value in E, if performed when the state is s. From this we can at once see more or less what it shall mean for such a theoretical model to *fit* a surface model. But not quite yet: it only tells us the probabilities of surface phenomena, on the supposition of a measurement *and* of a state. The latter is again something theoretical, behind the phenomena. A stringent notion of 'fitting' could go as follows:

> A theoretical model *MT fits* an experimental model *ME* just in case *MT* has some state s such that the function P_s^m contains the surface

state of *ME*, relative to the given identification of the measurement setups as measurement of the physical magnitudes *m*.

The situation is thus, so to speak, represented "from below" and "from above". In the surface model we have a representation that can be prepared in the laboratory or observatory, without recourse to the theory and its theoretical models.[21] But the theoretician has a look at the same situation 'from above', by specifying how it can be represented by a theoretical model. A way to evaluate what s/he provides along this line is presented by the above notion of *fit*. Let us look at a concrete illustration.

The surface model of an EPR experiment

As example I want to take a much discussed and described experiment that was inspired by the Einstein–Podolski–Rosen ("EPR") paradox, proposed by David Bohm, and eventually carried out by Alain Aspect. There is no better or more generally accessible presentation of this sort of experiment than the article David Mermin wrote for *The Journal of Philosophy* (1981). This article makes very clear that there are two, quite independent descriptions of the experiment, one 'from above' (in quantum theory terms) and one 'from below' that can be understood without knowledge of the theory.

As Mermin writes, it is not beyond present technology to mass produce the experimental set-up in a form to be sold in local drugstores and operated by anyone.[22] There are three unconnected devices, a transmitter and two detectors. The latter we may call L (for left) and R. Each has a dial with (say) three settings or orientations. When the detectors are turned on and the transmitter button is pushed, each detector flashes a light, which is either red or green. For ease of presentation, let's introduce some notation:

L1: the proposition that L has been given the first setting.

Lx: the proposition that L has been given the xth setting: $x = 1, 2, 3$.

The experiments have each two distinct possible outcomes, the red and green light flashes, which we may represent by the numbers zero and one:

L30: the proposition that L has the third setting and outcome zero.

Lx0: the proposition that L has the xth setting ($x = 1, 2, 3$) and outcome zero.

Lxa: the proposition that L has the xth setting and outcome a (a = 1, 0).

After a particular run on this apparatus, some of these propositions can receive a score of T (true) or F (false). Propositions about settings other than the actual one receive no score—they are irrelevant.

Suppose for example that for the first run, each apparatus was placed in the first setting; L had outcome 1 and R had outcome 0. An experimental report looks, in part, as follows:

Proposition	**Score**
L1	T
L2	F
R1	T
L10	F
L20	No score
R10	T

L20 received no score. Since it is the proposition that L was placed in setting 2 and had outcome 0, it could have been given F, simply on the basis that L2 was false. But L20 is useless information for it records an outcome for an arrangement that was not actualized, and so does not appear in the experimental report.

This single report is in any case not likely to come to the theoretician's desk. What reaches him rather is like:

(S) With initial preparation X, the probability of outcome Lxa, given setting Lx, equals **p**.

(L) For all initial preparations, the probability of (Lxa & Rxa), given settings Lx and Rx, equals zero.

which is an extrapolation of a summary of many reports of the above sort, and of the frequency counts made on their basis.

There is a gulf not only between the finite set of data points and the extrapolation to relative frequencies, but also between the latter and probabilities.

Accordingly there is a distinction between two intellectual artefacts that lie between the collection of specific, concrete results on the one hand and what the theorist takes as his or her input on the other. The reported relative frequencies, in their own summarized form constitute

the *data model* obtained from repeated applications of a single measurement procedure. What the theory confronts is abstracted from many data models. The abstracting is an idealizing, an extrapolation to a form that *could not* be reached in actual practice. So the reports (S) and (L) and their cognates, describe an idealized but not yet theoretical structure—the *surface model*.

Patrick Suppes pointed out that through construction of data models the experimentalist is in general bringing the theoretician small relational structures, constructed carefully from selected data. In the specific examples that Suppes mentions the little structures are algebras; hence he calls them *empirical algebras*.[23] Literature on the foundations of quantum mechanics typically points to such small structures that represent data, but they are not always algebras. They are more generally partial algebras, or just partially ordered sets ("posets") with some relations and/or operations. In the Appendix I will illustrate this, but the above already suffices to show how we need to think of the deliverances of experimentation and measurement as taking a much more general form than simple number assignments.

The over-arching concept for measurement

The bounds we set to the form that surface models can have, how we conceive that form, has changed historically. The bounds to the structure a logical space, supplied by a theory to delineate its models, can have, that has changed historically as well.

All the examples we have examined of measuring procedures are certainly cases of grading, in a generalized sense: they serve to classify items as in a certain respect greater, less, or equal. But as we also saw, this does not establish that the scale must be the real number continuum, nor even that the order is linear. The range may be an algebra, a lattice, or even more rudimentary, a poset. Generalizing on such examples, we can still conclude what I announced at the outset: *By measuring we assign the item a location in a logical space*.

So locating something in a logical space is the over-arching concept under which all actions of measurement can be arrayed. This is the only

stable stopping point we have found in the successive generalization of the notion of number-assigning.

Let's emphasize again that we cannot equate the two, however. Not every act even of locating something correctly in a logical space is the expression of an actual measurement outcome.[24] To say for example that our solar system is a Newtonian mechanical ten-body system, that is to locate our earth, the visible sun, and our sister planets in a logical space: the space of Newtonian models, which is the logical space of Newtonian physics. The bare assertion does not state a measurement result. But it is meaningful in part because it is not unconnected with what measurement can show. Can the assertion be taken as expressing a *possible* measurement outcome? In this case, certainly. A very large family of measurements normally so-called can be viewed as a 'battery', as a single complex measurement, and it can be so viewed within a certain theoretical context. In this complex outcome, our sun and planets 'look like' a Newtonian system. There are, however, more abstruse assertions within the language of contemporary physics that are much harder to see as thus connected with measurement, even *sub specie* the pertinent theory. There is certainly a general concern in the empirical sciences that the conditions of application of their concepts be closely connected with experimentally realizable conditions, but there is no rigid limit to how loose this "closely" can be.

Perspectival effects within logical space

While I do not want to stretch the term "perspective" very far beyond its narrowly literal meaning, there are places in this discussion of measurement where it seems apt enough. The aptness depends on context, and specifically on what theories count as accepted in that context.

If a thermometer is used by one person to locate the air in his room on the Fahrenheit scale, and by another to locate it in the space of possible mean kinetic energies of its molecules, the two are locating the same thing, by means of the same instrument, in two different logical spaces. That we should not at once call a change of perspective, though it certainly marks a change in ways of thinking about the air.

What if a theory then equates temperature and mean kinetic energy? In that case we should say that *relative to the theory* it is appropriate to call this a change of perspective. This qualification is important, though it may be left tacit in a context where the theory has been entirely accepted.

Quite a different case is posed by two different measurement procedures that are such as to assign always (error and vagueness aside) the same locations in the same space. As an example consider several methods for measuring velocity of a fluid, by pressure probe (use of pressure drop across a nozzle—Bernoulli effect), hot wire anemometer (heat transfer from a heated wire, due to convection heat transfer), and laser Doppler velocimetry (use of light scattering from small *seed* particles, Doppler shift). These can all be calibrated so as to give the same reading for the fluid velocity; there is no difference even in scaling as there is with the Fahrenheit and Celsius scales. It does not seem right to say here "there are differences in how things look from here", but rather "things look the same from here as from there". The moment we introduce this "from here and from there" distinction, however, we also have reason to warrant the *perspective* appellation.

What if two persons use different but similar thermometers to locate the same room's air on the Fahrenheit scale and Celsius scale respectively? We can't say that they use different methods of measurement, nor that they are measuring different parameters. Abstractly speaking we still have a choice: should we speak of two logical spaces, or of different coordinate systems in one space? But the latter choice is here so natural, that it almost seems forced on us. It would certainly not be outré to speak of a difference in perspective.

The two thermometers assign locations on two different scales, and these can in principle be taken as two logical spaces. But there is a transformation that systematically connects the readings on the one with the readings on the other. Given knowledge of that transformation, we immediately have a concept of a single logical space in which the two scales are, in effect, two coordinate systems. The transformation is linear; we immediately conceive of all the admissible temperature scales as connected to the Fahrenheit scale by linear transformations. With this way of looking at the matter, we are inclined to think of the use of Fahrenheit and Celsius thermometers providing different perspectives on the same magnitude, whose logical space is a more abstract structure.

We can see a similarity to the use of radar by two inertially moving traffic policemen, in motion relative to each other, to determine the speed of one and the same car. Their readings are different, for what they obtain is the car's speed relative to their own car—but whether they know it or not, there is a transformation that translates their different readings into one

another. The two policemen we say, surely, have before them the contents of two perspectives within the same space—literally in this case.

At the same time we should recognize that *relative to the pertinent theory* there is no real significance to a choice between a family of spaces related by a group of transformations and a family of coordinatizations of a single space. Little but bookkeeping ease is involved in the choice between speaking of different spaces, transformable one into another, and different coordinate systems imposed on a single space. Yet if we think in the former terms it seems less apt to speak of differences of perspective. Perhaps the resistance is the more reasonable, the more we recognize that there is high theory in play; and the one important thing is to pay attention to what theories are playing the background role.

Reconciling two views: perspective and invariance

> *For measurement the distinction is essential between the 'giving' of an object through individual exhibition on the one side, in conceptual ways on the other.* The latter is only possible relative to objects that must be immediately exhibited. That is why a *theory of relativity* is perforce always involved in measurement.[25]

This is Herman Weyl's rather cryptic elaboration on how measurement necessitates reference to something playing the role of a coordinate system associated with the perspective embodied in the measurement set-up. But there is a good deal of measurement in which this attributed relativity or perspectivity does not readily spring to the eye. What are we to make of measurements that yield outcomes ostensibly independent of differentiating features of the measurement arrangement?

Imagine a container with a movable piston, a pressure gauge and thermometer attached, and a scale on the outside that indicates the volume for the various positions of the piston. Three measurements are being carried out on this object, and together they locate its content in the three-dimensional Pressure-Volume-Temperature manifold—itself represented perhaps by the space of triples of non-negative real numbers. The outcome registered at some particular time, locates the container contents in that space.

This illustrates well the conclusion, that measurement is an action of locating the measured object in some logical space. But what has happened now to the much emphasized insight that a measurement outcome is after

all only a representation of the target, and in general does not show what that is like but only what it 'looks like' in that measurement set-up?

Let us honor these two views of what measurement does with the names *Measuring is Locating* and *Measurement is Perspectival*. Are they in tension with each other at all? One small point may help: what is perspectival is not the action of measuring but the contents of the measurement outcome, and locating is an action, not a content. Action and outcome are two different kinds of things. But this distinction does not go all that far: the measurement outcome does after all represent the target as located in a certain logical space. If we understand *Measuring is Locating* as meaning just that, we are back with two takes on the same thing, on the outcome.

If we are to call measurement perspectival, we need to qualify and elaborate if we are to arrive at an accurately made point. Let's begin with a classic passage in which Poincaré insists on the outcome's relativity to set-up:

> Lorentz could have accounted for the facts by supposing that the velocity of light is greater in the direction of the earth's motion than in the perpendicular direction. He preferred to admit that the velocity is the same in the two directions, but that bodies are smaller in the former than in the latter. If the surfaces of the waves of light had undergone the same deformations as material bodies, we should never have perceived the Lorentz–Fitzgerald deformation. In the one case as in the other, there can be no question of absolute magnitude, but of the measurement of that magnitude by means of some instrument. This instrument may be a yard-measure or the path traversed by light. It is only the relation of the magnitude to the instrument that we measure, and if this relation is altered, we have no means of knowing whether it is the magnitude or the instrument that has changed.[26]

"It is only the relation of the magnitude to the instrument that we measure"—that is either an overstatement or misleadingly put, though it emphasizes something important. The outcome of the measurement operation is a representation of the target, but it represents the object as it appears in that measurement set-up. However, it does not follow that the appearance there is different from the appearance of the same thing in other such set-ups. For after all there are invariants too! In the case Poincaré is discussing, the parameters measured are not invariant, they have different values in differently moving inertial frames. But there are parameters, even

ones definable from the results of local distance and time measurement outcomes, that have the same value in different frames.[27]

Note well though that the invariance we are now discussing is not the invariance that was cited as required for significance above. There we were concerned with the transformations that connect all the members of an admissible family of scales. Now we are discussing parameters for which the value registered in a measurement outcome is the same under admissible variations in the measurement set-up. As an example imagine again the speed of a car measured by radar from a moving police car. This speed can be registered on a scale of miles per hour or kilometers per minute, and so forth. Leaving aside data about the police car's own speed relative to the road surface, what is registered is the speed of the target relative to the radar source. If the policeman drives at a different speed himself, that relative speed of the target will be different: it is not an invariant. If on the other hand the radar is used to measure the target's acceleration, the result will be the same for any speed the police car may have. More precisely: where Newtonian mechanics applies, the acceleration is the same in all inertial frames, for all inertially moving measurement set-ups, while the speed is not.

There are therefore measurement outcomes that have no relativity left. Generally, these are after instrumental outputs have been processed with paper and pencil operations, with final outcome deduced relative to a theory. This is an important point: a measurement and its outcome can be complex, and include calculations and input from a model or theory. Such a procedure still fits the general idea of an operation performed so as to create a representation of the object; one that locates it in a certain logical space, with a location that it does not have a priori. To see how the activity signified by the slogan *Measuring as Locating* intertwines with the fact that *Measurement is Perspectival*, let us take a look at some examples of how 'simple' measurement outcomes are combined in such way to yield an outcome of a thus created 'complex' measurement.

Think once more of celestial navigation in the days of sailing ships. Navigating consists in locating oneself and guiding oneself from one location to another. The first part is the measurement whose outcome will govern that self-guidance.

What is one locating oneself in, in this case? In the grid, apparently first proposed by Hipparcus: basically our system of latitude and longitude which was also Ptolemy's. In Ptolemy's coordinate system as in ours, the

equator is at 0 and the North Pole at 90 degrees; similarly 360 degrees of longitude cover the earth, though 0 was not at Greenwich. For the ship's longitude to be determined, it would be sufficient to know what time it was at the same moment—e.g. local high noon—at 0 longitude, since the Earth rotates 15 degrees per hour. Precise time reckoning was a problem—let's leave this aside here.

To determine the latitude by means of local measurements was possible, however. The astrolabe that I mentioned before was succeeded in the eighteenth century by the sextant, to measure the elevation (angular altitude above the horizon) of celestial bodies. The elevation of the sun at noon, is a straightforward function of latitude. In the Northern hemisphere, the elevation of the North Star above the horizon is equal to the latitude. So imagine a sailing ship in the eighteenth century, with a sailor using a sextant to measure the elevation of the North Star. This places the North Star in a coordinatized Euclidean plane. Of course we do not read this as an "absolute" feature, it is its location relative to the observation set-up. The number found by the sailor is reported to the captain, who puts this to use to locate the ship on the earth. That is, he uses the reported number to locate the ship on the map, at a particular latitude—which together with his chronometer reading and his calculations of longitude yield a single point on that map.

Note the difference: at this point what for the sailor was the operation of finding the elevation *there* of the North Star has become for the captain a position measurement performed on the boat. The outcome of the complex measurement completed by the captain's manipulation and combination of data being the result of a deduction from that elevation value, relative to premises that the captain already had at his disposal, including some that are very theoretical. This measurement locates the boat in the geographic grid; it can be recorded without adding a "from here". The longitude, latitude, and time can now be entered in the log without any such 'relative' marker.

As a second example think of three cameras fixed on a grid, taking photos of an ancient coin. In each photo the object has an elliptical shape. In the simple set-up defined by one of the cameras, the shape measurement outcome is "elliptical, with axes . . .". But in the three-camera set-up, complemented with a little computer to process these three individual outcomes, there is a big measurement going on, with an outcome such as e.g. "circular". This too is a shape measurement. But here the outcome seems not to be relative to the cameras.

Yet this example brings out how context-dependent these judgments of relative/non-relative are. We might have to see something relative where we did not notice it before. In this example I probably tricked you into imagining this as an inertial frame in which objects and cameras are at rest with respect to each other. Shape is not a relativistic invariant, and if you think about it again in the new context that I have now introduced into our discourse, what was classified as not relative is reclassified as relative to something after all.

But all the shape measurements located the object in the same logical space of shapes. There is no contradiction when it is noted that the location of the object in that logical space is relative to the measurement set-up—and that adding in some other data and bits of theory will yield more complex measurement procedures in which invariants are measured.

Finally, single simple measurements may be combined into complex ones over time. Imagine a Stern–Gerlach apparatus with the usual two channels and screens at their ends. When the first black spot appears, and it is on the upper screen, we can say "looks like a particle in a definite spin-up state was prepared". After, say, a hundred spots one might say "looks like this source is preparing particles in a state in which 1/3 register as spin-up and 2/3 as spin-down". There is no falsehood here, nor any contradiction, for the two sorts of states mentioned can look the same in a single measurement. After a long run, the "looks like" can disappear: the outcome is that the source prepares particles in a superposition or mixture of spin-up and spin-down eigenstates . . ." etc. This can be the output of a program on a computer coupled to the apparatus; obviously certain assumptions or forms of inference are built into the algorithm that defines the program. The outcome in this case locates the source output with respect to a certain Hilbert space.

What is a measurement outcome?

We cannot distinguish what measurement is by attending solely to the physical correlate of measurement. The representational role is exhibited only if we attend to how the measurement outcome is a representation of what is measured.

A measurement is a physical interaction, set up by agents, in a way that allows them to gather information.[28] The outcome of a measurement

provides a representation of the entity (object, event, process) measured, selectively, by displaying values of some physical parameters that—according to the theory governing this context—characterize that object. It is a specific kind of representation, with some of its features inherited from the nature of representation in general and some others peculiar to it. The various strands in the way we think of measurement are to be disentangled—though without implying for a moment that the distinctions drawn are more than distinctions of reason, or that the different aspects we can distinguish could exist all on their own.

To disentangle some of the strands in the way we think of measurement, we can begin by emphasizing again the dual character, common to all that properly belongs under the heading of technology: that we describe the processes and objects involved *both in physical and in functional terms*.

What have we found so far? First of all there is the *relationality* in the concept of measurement outcome. An outcome is an outcome of something, of a process, which must be of a sort that satisfies various stringent conditions. This is already correct on the purely physical side. But however smoothly the story goes, of how certain implied correlations qualify a physical process as being usable for gathering information, there is still more to it. First of all, more is in play than a comparison of *amounts* of information transmitted, and more than a correlation between physical states of object and apparatus. The measurement was *designed to answer specific questions*, and the information provided by the outcome is *relevant* to their answer. Notions such as relevance and reference, as well as the questions that define the context, are at best assumed as given when a technical measure of the amount of information is applied.

The information borne by a measurement outcome refers to a specific object or process (what was measured) and is selective with respect to characteristics of that referent. Such very simple examples as, the information provided by a reading of a thermometer (concerning the ambient temperature) or a pressure gauge (concerning tire pressure), may suggest that information transmission is something that can be explicated in purely physical terms.[29] But if a physical item is classified as the outcome of a measurement (of a specific kind) then it is classified as a representation, and therefore in effect classified under an intensional concept. The outcome provides a representation ***of*** the measured item, but also represents it ***as*** thus or so.

We can see this as soon as the phrase "carries the information that . . ." is employed. To adapt an old example from Frege: to carry the information that the Evening Star is within 15 degrees of the sun is not to carry the information that the Morning Star is thus, although these are the same object, and although therefore correlation with the position of the Evening Star is automatically correlation with the position of the Morning Star.

The *intensionality* of the concept of a measurement outcome consists in the fact that it is something that has meaning. In reporting the outcome one says, for example, that the pressure was [found to be] 17 psi; that report is a sentence expressing a proposition. Even if, given the background knowledge or opinion about the whole set-up, the pressure was necessarily 17 psi if and only if $17 = rT/V$, the outcome was *not* that the value of rT/V was 17—not even in the context defined by that background knowledge or opinion.[30]

That the concept is intensional is not to be confused with its being *intentional*. Literally, "intentional" refers to intention, but we take it broadly to include purpose, goal, role, and function. To classify something as a measurement outcome is to classify it as playing a certain role, namely as the outcome of a process with a definite *function*. This is entirely in line with the reflection that the activity of measurement belongs to technology, and technological concepts have this dual character, of referring to physical entities but partly (and essentially so) in terms of function.[31]

Fourthly, we must insist on the *indexicality* of the measurement outcome content in general. That is easy enough to spot when the measurement outcome is the indexical proposition that the iceberg is located 17 leagues to the North-East. But it is an especially significant feature when the context is more, rather than less, theory-laden.

Suppose for a moment that I take a pressure reading on the tire of my car. The outcome can be reported simply and precisely as attributing a feature to that tire. Where is the indexicality in that? But think of how different a role this report plays from the assertion, written in the very same words, in a historical account of what someone or other did somewhere. For this outcome to be something useful for me, I must appreciate that by means of this measurement operation I have also located my own situation—which involves *this* tire—as having a place in the theoretical region of pressure-graded objects.

Although this sort of representation is indexical in the same sense that 'first person' discourse is indexical (cf. Perry 1993), it is not 'subjective' in contrast to 'public' or 'inter-subjective'. Measurement outcomes are public and they are intersubjectively accessible; this is crucial to the methodological requirement of reproducibility in scientific experimentation. The measurement outcome content typically involves indexical reference to a particular vantage point, but this vantage point is a publicly ascertainable feature of the measurement set-up.

This fourth characteristic, the indexicality, is one not shared by representation in general. It derives from the specific function which this particular method of representation has.

There are further features that relate the measurement outcome to specific kinds of representation, notably to imaging and picturing. Recall that by *imaging* I mean representation that trades on selective resemblances, and that the special case of *picturing* was introduced above as follows:

> Overlapping these categories of representation that trade on selective resemblances lies still a further salient case, which shares some crucial features found in visual perspective, a development which in art we associate specifically with Renaissance painting. Perspective involves . . . such features as *occlusion, marginal distortion, texture-fading*. For cases of imagery in which such features of perspective are present I'll use the terms *picture* and *picturing*—these can include cases of kinematic and visual but perhaps also still further forms of imagery.

The most easily recognized cases of picturing are ones in which the resemblance is not at a very high or abstract level: it is just a sharing of selected properties. The subway map shares the topological structure of the subway system, say; it is a picture of that system. But resemblance, we recall, can be higher order: the spatial structure of a set of letters on a page may be the same as the temporal structure of a set of events named by those letters. The use of visual or kinematic imagery to depict things that are not visual or kinematic is rife, and not excluded by our notion of imagery. So a measurement outcome may well purport to give us information by means of a selective resemblance to what is measured, although the pointer-observable may be of a very different character than the measured observable.

Indeed, the Criterion for the Physical Correlate of Measurement entails that *resemblance at such a structural level* is *required* to be implied by the

theoretical classification of something as a measurement. So this, the trading on selective resemblance at some level, is our *fifth* characteristic.

And finally, there is the sense in which measurement outcomes are perspectival. This has been explored extensively by Ronald Giere (2004; 2006), whose work has placed this topic at center stage.

There is historic precedent. Poincaré (1897: ch. 1) emphasized strongly the 'perspectival' ('relational' in the sense of *relative* rather than *derived*) character of the outcome. Poincaré overstated it perhaps ("It is only the relation of the magnitude to the instrument that we measure"), but the point is clear. What the outcome reveals is not directly what the measured object is like, but what it 'looks like' in that measurement set-up. The latter point was exploited saliently by Einstein in his 1905 STR paper, where measurements of time and of length are taken to yield only data explicitly relative to the state of motion of the measurement set-up.[32]

To assess this question of perspectivity properly we must take stock of its hallmarks, such as *occlusion, marginal distortion,* and *texture-fading*. Literally, marginal distortion is distortion in proportions toward the outer edges of a perspectival drawing or painting. In a more general sense, marginal distortion is distortion that is the result of the limits inherent in a given mode of representation. Such would seem to be a feature of almost any sort of measurement at all, given that instruments have a limited range, and become inevitably less reliable and less exact near the limits of that range. A liquid thermometer is not to be used near the freezing and boiling points of that liquid, for example. Perhaps a simple detection of presence or absence, with a yes–no answer, would be a counterexample, but then only if the means of detection could not possibly give uncertain results—an unlikely case. In the context of quantum mechanics we have a more extreme sort of example. That an object's location may, as far as its state is concerned, be probabilistically smeared out over the whole of space, while we can only get our measuring apparatus to ask whether or not it is in this room (or similarly small finite region), means that information gatherable beyond a narrow range is maximally indefinite—that is marginal distortion.

Occlusion and texture-fading are easy to spot in the sort of measurement that is paradigmatic for our concept, namely the sort of spatial measurement I discussed with reference to Hugh of St. Victor, Alberti, Brunelleschi, and Dürer.

It is tempting to see these same concepts as applying in quantum mechanics due to the impossibility of simultaneous measurement of non-commuting observables. If one observable is measured, the extent of information that can be gathered with respect to incompatible observables is drastically reined in: that remind us of occlusion. That the observable really measured, when a measurement is officially designed to measure a sharp observable, will necessarily be a fuzzy version of that observable—that sounds like texture-fading. But to so use the terms is at best an analogical extension of their literal meaning. Occlusion means *hiding*—changing something so that it can't be measured is therefore not occlusion.[33] In the case of quantum mechanics it is certainly not the case, for example, that the measured value of momentum implies a real but hidden value of position. On the other hand, if a system is in an eigenstate of an observable, and an incompatible observable is measured, then that state has changed. So we have two reasons not to call this occlusion. As to texture-fading: the fuzzying discussed for quantum measurement does not have anything to do with distance, which was what concerns texture-fading in pictures. But though the analogy is far from complete, it is not far-fetched to analogically extend the notions of occlusion and fading in this way.

Relating the views 'from above' and 'from within'

As long as we consider what happens in measurement purely from the theoretical point of view, the only criteria are theory-internal. The criteria of adequacy cannot go beyond coherence. That is why the theoretical point of view remains empty unless the problem of coordination is also faced and taken into account. "Concepts without percepts are empty"—this Kantian motto transposes to the empiricist point that theory without coordination is empty. To understand how theoretical parameters become coordinated, we have to look into the historical process in which measurement procedures and theory evolve together, in a thoroughly entangled way. Nor can this evolution start in anything but *prior* meaningful discourse relating to what *eventually* will be delineated as the theory's domain.

The scientific realist will interpret this process, in which practice and theory jointly stabilize, as establishing that the theory latches on to the blueprint for the universe.[34] But the one empirical fact is precisely this:

that the practice, both experimental and theoretical, stabilizes, and that 'nature cooperates' to the extent that, perhaps temporarily, no or less resistance is experienced in this practice.[35] For the time being, at least, the expectations engendered by empirical predictions are satisfied, the retrospective evaluation says that thus guided empirical judgment has been well calibrated.